

Hilbert Geometry at Surreal Scales

Orthogonal splitting, amplified and infinitesimal graphs,
projection-lattice obstructions, metric rigidity,
and Fredholm least squares

Built from two manuscripts dated September 23, 2026

Abstract

We develop Hilbert geometry over set-sized surreal and surcomplex Hahn fields while retaining an ordinary Hilbert space at each coefficient. The scalar-valued inner product is positive definite, its norm exists without a divisible value group, and the valued space is spherically complete. These facts do not imply the projection theorem.

Using the collection’s automatic adjoint theorem, every orthogonally complemented subspace is an infinitesimal unitary deformation of an ordinary closed subspace, equivalently the graph of a positive-order Hahn series of bounded operators over its residue. For an ordinary bounded T and an infinite Hahn scalar a , $\text{Graph}(aT)$ is complemented if and only if $\text{ran } T$ is closed; for an ordinary closed densely defined A and a positive infinitesimal q , the graph of qA is orthoclosed and spherically complete but complemented if and only if A is bounded. Both are instances of one criterion: an infinitesimally scaled closed linear relation is complemented exactly when its domain is closed. For a nontrivial value group the poset of complemented subspaces is a lattice exactly when the coefficient space is finite dimensional; a positive operator with three exponents has a noncomplemented kernel, and a countable orthogonal family of rank-one projections has no join. A positive Hahn metric with strictly positive residue is isometric to the canonical one exactly when its residue is uniformly positive.

A positive counterpart: every positive-order Hahn perturbation of an ordinary Fredholm operator has an adjointable Moore–Penrose inverse, and a finite Schur matrix determines its kernel, cokernel, index and exact inverse valuation. Adjointable operators on the class of all surcomplex vectors have one set-sized support.

This report merges two independently written manuscripts; it says which source proved each result, prints shared results once, and keeps every source’s limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources and status. Manuscript 03 (the base text), pinned to repository commit [9a385d3](#), and manuscript 06, pinned to [a45d722](#); their titles are in [Section 1.4](#). Results already printed elsewhere in the collection are cited, not reprinted. The construction, the automatic adjoint theorem, the direct-rotation formula and the classical perturbation tools are credited, not claimed. The provenance of every section is recorded in [Section A](#); non-claims are collected in [Section B](#). Finite checks accompany the report and check finite identities only. No named published conjecture is asserted to be solved.

Contents

1	The question, the model, and the results	1
1.1	The model	1
1.2	Main results	1
1.3	Conventions and renamed symbols	2
1.4	Provenance of this report	3
2	Scalars, supports, and the surreal interpretation	4
2.1	Standing conventions	4
2.2	Actual surreal and surcomplex components	5
2.3	Why the whole surreal class is not silently a set field	6
3	The Hahn–Hilbert space and its completeness	6
3.1	The inner product and its norm	6
3.2	Valuation topology and spherical completeness	7
3.3	Finite-dimensional Hilbert geometry survives	8
4	Coordinates, two summation levels, and duality	8
4.1	An exact coordinate-space description	8
4.2	What becomes of an orthonormal basis	9
4.3	Riesz representation requires more than valuation continuity	9
4.4	Least field-valued norm bounds	10
5	The intrinsic algebra of adjointable operators	10
5.1	Operators with a common Hahn support	10
5.2	Positive-order operator calculus	12
5.3	Finite-rank operators and self-adjoint inverses	12
6	Orthogonal splitting and its residue normal form	12
6.1	Three equivalent formulations	12
6.2	Every projection has ordinary residue	13
6.3	A global normal form using the classical direct rotation	14
6.4	Exact graph coordinates for a fixed residue	14
7	Unitary structure and finite simultaneous straightening	16
8	Amplified and infinitesimal graphs	17
8.1	The normal equation is the whole obstruction	17
8.2	Finite and infinite scalar amplification of bounded operators	17
8.3	Infinitesimal graphs of closed unbounded operators	19

8.4	The common criterion: scaled closed linear relations	21
9	The projection poset: meets, kernels and countable joins	22
9.1	What survives: orthocomplements and orthogonal sums	22
9.2	Two good subspaces with a bad intersection	23
9.3	A positive operator with a noncomplemented kernel	24
9.4	A mutually orthogonal countable family with no join	24
10	Metric rigidity and the exact positivity threshold	25
11	Fredholm perturbations reduce to a finite defect matrix	27
11.1	The hypotheses and the reduction	27
11.2	Finite Moore–Penrose geometry	29
12	Exact Moore–Penrose inverses and least squares	29
12.1	Finite defects restore orthogonal range decomposition	29
13	The exact amplification scale from finite minors	30
13.1	Finite valuation-ring reduction	30
13.2	A sharp formula for the Moore–Penrose inverse	31
13.3	An off-diagonal term can dominate the inverse scale	33
14	Coherent extension of scales and the full surreal class	33
14.1	Set-sized scalar extension preserves the formulas	33
14.2	A literal vector space over all surcomplex scalars	34
14.3	Global fine topology: set-sized discreteness	36
15	The report in the collection, the literature, and open questions	37
15.1	Relation to neighbouring reports	37
15.2	Primary literature and the novelty boundary	38
15.3	Open questions	39
15.4	Proof status	40
A	Provenance	41
A.1	Section by section	41
A.2	Labels	42
A.3	Renamed symbols	42
A.4	Stale and incomplete repository statements	43
B	What this report does not claim	44
B.1	Source 03 (34 items)	44
B.2	Source 06 (33 items)	45

B.3 The merge (6 items)	47
C A dependency and hypothesis checklist	48
D Verification suites	49

1 The question, the model, and the results

Replacing real or complex vector components by surreal or surcomplex numbers is not merely a change of notation. In infinite dimension one must choose which coordinate families are vectors, which infinite sums are legitimate, which topology defines completeness, and which operators are admitted. Different choices produce genuinely different theories. This report makes one such choice and proves both its strengths and its unavoidable limitations.

1.1 The model

We use the coefficient-Hilbert model already constructed in Part I of the collection's report *Infinite-Dimensional Hahn Spectral Theory* (directory `docs/surcomplex/infinite-dimensional-hahn-spectral-theory`, label prefix `ih`s:) [17]:

$$\mathcal{H} = \mathcal{H}_\Gamma(H) = H((t^\Gamma)), \quad F = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)) = F[i].$$

Here H is an ordinary Hilbert space and Γ is a set-sized ordered abelian group, nonzero unless the contrary is said. A vector is a formal series of ordinary Hilbert vectors with well-ordered support. Through a chosen normal-form embedding of the scalar field, its coordinates are actual surcomplex numbers. There is also an entirely real version, with H real and F in place of K ; except where said otherwise the text is written in the complex case and every statement and proof holds verbatim in the real case. Its duality and completion are developed in the collection's report *Three Duals at Surreal Scales* (prefix `du`als:) [18].

This model retains every ordinary vector of H , not only vectors with finite coordinate support. It also retains ordinary bounded operators as constant coefficients. It does *not* retain ordinary Hilbert norm convergence as the topology on the constant copy of H . The distinction is crucial both for what works and for what fails.

1.2 Main results

Call a K -linear subspace $M \subseteq \mathcal{H}$ *orthogonally split*, or *orthogonally complemented*, when $\mathcal{H} = M \oplus M^\perp$. This is equivalent to every vector having a nearest point in M , and to M being the range of a self-adjoint idempotent (Proposition 6.2).

- (1) *Classification of complemented subspaces* [03, 06]. Every split M is $M = U\mathcal{H}_\Gamma(M_0)$ for a unitary U with $v(U - I) > 0$ and a unique ordinary closed $M_0 = \text{res } M$; equivalently it is the graph of a positive-order Hahn series in $\mathcal{B}_\Gamma(M_0, M_0^\perp)$ (Theorem 6.5 and Corollary 6.6). Given the collection's rotation lemma `ih`s:fr:lem:rotation, this is a corollary of an integrality lemma for projections (Lemma 6.3). Closed residue is necessary but not sufficient (Proposition 6.7).
- (2) *Unitaries* [06]. Unitary orbits of projections are classified by two ordinary dimensions (Corollary 7.1); every adjointable unitary is $\widehat{U}_0 \exp X$ uniquely (Theorem 7.2); finite orthogonal decompositions straighten simultaneously (Theorem 7.3).
- (3) *Two graph theorems and their common criterion* [03], [06], [merge]. For ordinary bounded $T : H \rightarrow G$ and $a \in K$,

$$\text{Graph}(aT) \text{ is split} \iff a = 0 \text{ or } v(a) \geq 0 \text{ or } \text{ran } T \text{ is closed,}$$

and for $v(a) < 0$ the residue is $\ker T \oplus \text{ran } T$ (Theorem 8.3). For ordinary closed densely defined A and a positive infinitesimal monomial $q = t^\eta$, the graph $M_q(A)$ of qA is orthoclosed

and spherically complete, with residue $\text{Dom } A \oplus 0$, and is split if and only if A is bounded (Theorem 8.7). Neither theorem implies the other; both splitting criteria follow from Theorem 8.10: for a closed linear relation R and $v(b) > 0$, the scaled relation $M_b(R)$ has residue $\text{dom } R \oplus \text{mul } R$ and is split exactly when $\text{dom } R$ is closed.

- (4) *The projection poset* [03, 06]. For $\Gamma \neq 0$, $\text{Spl}(\mathcal{H}_\Gamma(H))$ is a lattice if and only if $\dim H < \infty$ (Theorem 9.2); a positive operator with exponents $0, \eta, 2\eta$ has a noncomplemented kernel (Theorem 9.4); a countable family of mutually orthogonal rank-one projections has no least upper bound, and countable simultaneous straightening fails for two distinct reasons (Theorem 9.5, Corollary 9.6, and Proposition 9.7).
- (5) *Metric rigidity* [06]. A positive metric $\Theta = \widehat{\Theta}_0 + \Theta_+$ with strictly positive residue is isometric to the canonical one by some linear bijection, with no regularity assumed, exactly when Θ_0 is uniformly positive (Theorem 10.1).
- (6) *Fredholm least squares* [03]. For $A = T_0 + E$ with T_0 ordinary Fredholm and $v(E) > 0$, with $m = \dim \ker T_0$, $n = \dim \ker T_0^*$ and r the rank of a finite Schur matrix S ,

$$\dim_K \ker A = m - r, \quad \dim_K(\mathcal{G}/\text{ran } A) = n - r, \quad \text{ind}_K A = m - n,$$

$A^\dagger$ exists in the adjointable Hahn algebra, and for $r > 0$

$$v(A^\dagger) = -(\nu_r(S) - \nu_{r-1}(S)), \quad \nu_0(S) = 0,$$

where ν_j is the least valuation of a nonzero j -minor (Theorems 11.1, 12.1 and 13.2). No Archimedean hypothesis, divisibility, analytic convergence or bound on all coefficient operator norms is imposed. These theorems contain the determinant-free clauses of the collection's Hahn Fredholm alternative `ihs:fr:thm:fredholm` (Remark 11.2).

- (7) *The whole surreal class* [03, 06]. Working with all of **No** or **No**[i] in a class theory without global choice, every class-linear map with an everywhere-defined class adjoint has one set-sized Hahn expansion in ordinary bounded operators, so the whole theory applies to literal surreal and surcomplex vector components (Theorem 14.2 and Corollary 14.3).

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. The sources name their own candidate contributions and credit their inputs; the literature comparisons were targeted and priority is not certified (Section 15, Section B).

1.3 Conventions and renamed symbols

The two sources wrote the same objects with different letters, and some of their letters collide with the collection's usage. The following convention is fixed for the whole report; Table 3 lists every renaming.

- $F = \mathbb{R}((t^\Gamma))$ is the ordered real Hahn field and $K = \mathbb{C}((t^\Gamma)) = F[i]$ the complex one, as in source 03, in the spectral report (`ihs:`, where F is written `Fld`) and in the finite spectral-theory report (F_Γ , K_Γ). Source 06 had these two letters swapped (K real, F complex); they are swapped back throughout. $\mathbb{T} \in \{\mathbb{R}, \mathbb{C}\}$ is the coefficient field of H .
- $\mathcal{H}_\Gamma(H) = H((t^\Gamma))$ is the whole Hahn–Hilbert space. Source 06 wrote $\mathcal{E}_\Gamma(H)$; that letter is not used here, because in the three-duals report $E_\Gamma(V) = \text{span}_K V \cong K \otimes V$ is the *algebraic* scalar extension, in general a proper subspace of $V((t^\Gamma))$.

- $\mathcal{B}_\Gamma(H, G) = \mathcal{B}(H, G)((t^\Gamma))$ (source 06: $\mathcal{A}_\Gamma(H, G)$; the spectral report writes $\mathcal{A}_\Gamma(H)$ for $\mathcal{B}_\Gamma(H, H)$), $v(A) = \min \text{supp } A$ (06: v_{op}), and $\mathcal{B}_\Gamma^{>0}(H, G)$ is the set of series supported in $\Gamma_{>0}$, zero included. A hat denotes the constant lift of an ordinary operator.
- *Split* and *orthogonally complemented* are synonyms; a *projection* is an everywhere-defined self-adjoint idempotent (06: “projector”). *Closed* means valuation closed; *orthoclosed* means $M = M^{\perp\perp}$. The two are never used interchangeably.
- $q = t^\eta$ with $\eta \in \Gamma_{>0}$ is a fixed positive infinitesimal monomial (06: $\tau = t^\delta$).
- On $\ell^2 = \ell^2(\mathbb{N}_{\geq 1})$, $T = \text{diag}(1/n)$ (06: S ; the spectral report writes D), $\Lambda = T^{-1} = \text{diag}(n)$ with $\text{Dom } \Lambda = \text{ran } T$ (06: D), and $h = (1/n)_{n \geq 1} \in \ell^2 \setminus \text{ran } T$.
- $U(H)$ is the ordinary unitary group and $U(\mathcal{H})$ the group of adjointable unitaries (06: $\mathcal{U}$; the spectral report uses $\mathcal{U}(T)$ for the set of norm bounds of T).
- The Fredholm blocks and auxiliary operators of source 03 are renamed where they met the field letters or the rotation: see Table 3.

Inner products are linear in the first argument. The valuation of a vector or an operator is the least exponent of its support; $v(0) = +\infty$.

1.4 Provenance of this report

This report is built from two manuscripts written independently and dated September 23, 2026, items 03 and 06 of the twenty-seventh batch placed in commit a4dcb91; their shipped file prefixes are 03-orthogonal-splitting- and 06-hilbert-foundations-. Source 03 pins repository commit 9a385d3 (35 commits before that placement) and source 06 pins a45d722 (30 commits before it). None of the reports cited below changed between the pins and the placement.

Source	Manuscript	Principal contribution to this report
03	<i>Hilbert Geometry at Surreal Scales: Orthogonal Splitting, Amplified Graphs, and Fredholm Least Squares</i> (29 pp.)	Base text and structure; the residue normal form and graph chart; the amplification threshold for bounded operators; the lattice dichotomy with its finite-dimensional converse; the Fredholm, Moore–Penrose and inverse-valuation theorems; scale coherence; the class descent in NBG without global choice; fine-net stabilization.
06	<i>Hilbert Geometry at Surreal Scales: Orthogonal subspaces, infinitesimal graphs, metric rigidity, and projection-lattice obstructions</i> (25 pp.)	The c_{00} example; the strict-contraction functional; the classification in its own form, with the unitary-orbit corollary; the unitary group and finite simultaneous straightening; the closed densely defined graph theorem and the distance cut; the positive operator with a noncomplemented kernel; the missing countable join and the two obstructions to countable straightening; metric rigidity; the global geometry corollary; the Solèr framing.

Where the merge had to choose.

- *Base*. Source 03 is the base, because on every shared theorem it has the weaker hypotheses: the lattice theorem for every infinite-dimensional H with the finite-dimensional converse (06:

one direction, for ℓ^2), the class descent without global choice (06: with global choice), and an arbitrary amplification scalar (06: a monomial). The straightening theorem is identical in both.

- *Printed once.* The positive form, spherical completeness, automatic adjoints, the integral-projection lemma, the normal form, the graph chart, the no-meet construction (03's and 06's are the same up to swapping coordinates), the class descent and fine discreteness are proved by both sources and printed once, crediting both. Two genuinely different proofs of nonnegativity of projections are both kept (Lemma 6.3).
- *Printed by citation.* The automatic adjoint theorem and its bound (`ihs:hh:thm:adjoint`, `ihs:hh:cor:autobounded`), reproved by both sources, are cited; the rectangular form is obtained from the square one by a two-line reduction [merge]. So are the norm-attainment criterion (`ihs:hh:thm:norm`), the coefficientwise Riesz criterion (`duals:thm:riesz`), the direct rotation (`ihs:fr:lem:rotation`), the finite Moore–Penrose and determinantal-scale theorems of the finite spectral-theory report (`thm:leastssquares`, `thm:scales`, `lem:DeltaInvariant`), and fine discreteness (`a:thm:discrete`). Source proofs are kept only where they prove more or proceed differently.
- *The two graph theorems* are related but neither implies the other: 03's concerns bounded operators with arbitrary kernel and range and also finite scalars; 06's concerns unbounded operators and also computes the orthogonal complement. Both are printed with their proofs, and their common splitting criterion is added by the merge with a complete proof (Section 8.4).

Results added by the merge carry the tag [merge] and have complete proofs. Section A records, section by section, which source proved what; Section A.4 lists the repository statements of the sources that were corrected or completed.

2 Scalars, supports, and the surreal interpretation

2.1 Standing conventions

An ordered abelian group is written additively. A subset is well ordered in its given order, not in the reverse order. For a vector space V over $\mathbb{T}$,

$$V((t^\Gamma)) = \left\{ x = \sum_{\gamma \in \Gamma} x_\gamma t^\gamma : x_\gamma \in V, \text{ supp } x \text{ well ordered} \right\}.$$

Set $v(0) = +\infty$ and $v(x) = \min \text{supp } x$ otherwise. The exponent order is the valuation order: positive exponents are small. The field F is ordered by its leading nonzero coefficient, and complex conjugation on K acts on coefficients and fixes monomials. The valuation ring and its maximal ideal are

$$\mathcal{O} = \{a \in K : v(a) \geq 0\}, \quad \mathfrak{m} = \{a \in K : v(a) > 0\}.$$

Zero is included. Residue is extraction of the coefficient of t^0 ; on finite real scalars it is the standard part. A scalar is finite in the ordered-field modulus precisely when it is in $\mathcal{O}$. An infinite scalar has negative valuation.

Lemma 2.1 (Hahn–Neumann support calculus [06]). *If S, S' are well-ordered subsets of an ordered abelian group, then $S + S'$ is well ordered and, for every α , the set $\{(s, s') \in S \times S' : s + s' = \alpha\}$ is finite. If $S \subseteq \Gamma_{>0}$ is well ordered, its generated additive monoid S^* is well ordered and each*

exponent has only finitely many representations by words in S , counting all word lengths, apart from the unique empty word at zero.

The second assertion is the positive-support form of Neumann’s lemma [12]; with the first it is the spectral report’s support lemma `ihs:lem:support`, where a support-engine proof and attribution are given [17]. It is valid without an Archimedean hypothesis. For the first finiteness statement, infinitely many distinct first coordinates would have a strictly increasing subsequence, forcing a strictly decreasing subsequence of second coordinates, contrary to well-ordering. The same descending-sequence argument gives well-ordering of the sum set. The positive-word assertion is the classical support theorem, imported rather than inferred from convergence.

Definition 2.2 (Strong Hahn summation). A set-indexed family $(x_j)_{j \in J}$ in $V((t^\Gamma))$ is strongly Hahn summable if the union of its supports is well ordered and, at each exponent, only finitely many members have a nonzero coefficient. The sum $\sum_j^H x_j$ is formed by these finite coefficient sums. A linear map is *strong* if it preserves strongly summable families and their sums.

This is not the strong operator topology and not ordinary convergence. Modern treatments of formal summability and strongly linear maps include [3, 6]; the distinction is central in the three-duals report. Lemma 2.1 justifies every formal multiplication and positive-order power series used below. In particular, in a unital coefficient algebra,

$$(I + B)^{-1} = \sum_{j \geq 0}^H (-B)^j \quad (v(B) > 0). \quad (2.1)$$

For noncommutative coefficient rings this Neumann inverse is the spectral report’s `ihs:lem:neumann`, whose Lean formalization is recorded in `docs/FORMALIZATION.md`. Binomial series $(I + B)^c$ for real c , and e^B , $\log(I + B)$, are also defined by Lemma 2.1 (`ihs:cor:binomial`; see Section 5.2). These are identities of Hahn series. They do not presuppose topological convergence of the finite partial sums.

Example 2.3 (Summable families not converging through finite sums [03, 06]). Two families show the phenomenon.

- (a) The family $t^{n/(n+1)}$, $n \geq 1$, in $\mathbb{C}((t^\mathbb{Q}))$ is strongly summable. Its exponents increase to 1. The tail of every finite partial sum still has valuation below 1, so the partial sums do not converge in the valuation topology [03].
- (b) Let $\Gamma = \mathbb{R} \times \mathbb{R}$ with lexicographic order and $\delta = (0, 1)$. The strong sum $\sum_{n \geq 0} t^{n\delta}$ exists and equals $(1 - t^\delta)^{-1}$. Its finite partial sums have errors of valuation $(0, N + 1)$, which never exceeds $(1, 0)$, so they do not converge either. None of the positive-order operator power series below is justified by assuming $n\delta \rightarrow +\infty$ [06].

This distinction is already emphasized in the collection’s foundations and duality reports [21, 18].

2.2 Actual surreal and surcomplex components

Fix an order-preserving additive embedding $\iota : \Gamma \hookrightarrow (\mathbf{No}, +)$. For example, the usual $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ can be embedded using their standard surreal copies. Conway normal forms give the embedding

$$\sum_{\gamma} r_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} r_{\gamma} \omega^{-\iota(\gamma)}. \quad (2.2)$$

The support on the right is reverse well ordered in its exponents, exactly as a surreal normal form requires. The right-hand sum is a surreal normal form, not the limit of its finite partial sums in the full fine topology. Addition, multiplication, order, and strong summation are preserved; adjoining i gives an embedding into $\mathbf{No}[i]$ [5, 21]. If Γ is divisible, F is real closed and K algebraically closed; this familiar fact is used only where it is mentioned, and the norm construction below needs less.

For a fixed ordinary orthonormal basis (e_j) of H , every vector in $\mathcal{H}_\Gamma(H)$ therefore has coordinates in a specified subfield of $\mathbf{No}[i]$. Section 4 gives the exact coordinate admissibility condition.

2.3 Why the whole surreal class is not silently a set field

A nonzero vector space over all of $\mathbf{No}$ or $\mathbf{No}[i]$ cannot have a set-sized underlying collection: multiplication of any fixed nonzero vector by all scalars would inject a proper class into that set. Thus a literal global theory requires class-sized objects or a universe-relative formulation. The main theorems below are set-sized and can be transported by (2.2); Section 14 states a separate class-theoretic extension.

For any specified set of scalar normal forms, their supports lie in a common set-sized additive subgroup of surreal exponents. Thus particular calculations take place in a common Hahn workspace. This does not mean that every class-sized problem has been reduced to a single set field, nor that the full surreal fine topology equals the intrinsic topology of that workspace.

3 The Hahn–Hilbert space and its completeness

3.1 The inner product and its norm

For $x, y \in \mathcal{H}_\Gamma(H)$ put

$$\langle x, y \rangle = \sum_{\alpha \in \Gamma} \left(\sum_{\beta + \gamma = \alpha} \langle x_\beta, y_\gamma \rangle_H \right) t^\alpha. \quad (3.1)$$

At each exponent only finitely many coefficient pairs contribute (Lemma 2.1). The ordinary Hilbert inner products are evaluated *before* the Hahn convolution, possibly as ordinary infinite Hilbert sums; these are two different levels of summation. This is the spectral report’s coefficient-Hilbert inner product.

Proposition 3.1 (Positive-definite Hahn extension [03, 06]). *The form (3.1) is Hermitian and positive definite. If $x \neq 0$, $\delta = v(x)$ and $u = x_\delta$, then*

$$\langle x, x \rangle = t^{2\delta} \|u\|_H^2 (1 + r), \quad r = 0 \text{ or } v(r) > 0,$$

and $\langle x, x \rangle$ has a unique positive square root in F , regardless of whether Γ is divisible. With $\|x\| = \sqrt{\langle x, x \rangle}$,

$$v(\|x\|) = v(x), \quad v(\langle x, x \rangle) = 2v(x) \quad (x \neq 0).$$

Cauchy–Schwarz, the triangle inequality, the parallelogram identity, and finite Pythagoras hold over F .

This is `ihshh:prop:inner` of the spectral report (and `duals:prop:norm` of the three-duals report); both sources reprove it. For the reader’s convenience: the only pair contributing at 2δ is (δ, δ) , so the leading coefficient is $\|u\|_H^2 > 0$; the root is $t^\delta \|u\|_H \sum_{j \geq 0}^H \binom{1/2}{j} r^j$, a strong sum by Lemma 2.1 whose square is $\langle x, x \rangle$ by the binomial identity; uniqueness follows from $(a - b)(a + b) = 0$

in an ordered field. Expanding the nonnegative $\langle x - by, x - by \rangle$ with $b = \langle x, y \rangle / \langle y, y \rangle$ gives Cauchy–Schwarz, and the triangle inequality follows. No supremum or limiting argument is used. The same construction gives $|a| = \sqrt{a\bar{a}}$ for $a \in K$.

Warning 3.2. The assertion is not that every positive element of F has a square root. For example, t has no square root when $\Gamma = \mathbb{Z}$. Squared norms have even leading valuation, which is exactly why all roots needed here do exist.

3.2 Valuation topology and spherical completeness

A neighborhood basis of 0 is $\{x : v(x) > \gamma\}$, $\gamma \in \Gamma$. It is the same topology as that generated by all positive F -valued norm radii: if $v(x) > v(r)$ then $\|x\| < r$, while a monomial radius of exponent strictly greater than γ forces $v(x) > \gamma$. In rank one, when $\Gamma \subseteq \mathbb{R}$, $d(x, y) = \exp(-v(x - y))$, with $d(x, x) = 0$, is a real-valued ultrametric for the same topology. It is not the F -valued Hilbert norm itself. This is the topology of a fixed workspace, not the topology induced by all positive tolerances in the full surreal field ([Proposition 14.4](#)).

Theorem 3.3 (Spherical completeness of the coefficient space [03, 06]). *For every $\mathbb{T}$ -vector space V , with no completeness or topology assumed on V , every nonempty nested set of balls*

$$B(a, \gamma) = \{x : v(x - a) \geq \gamma\} \quad (a \in V((t^\Gamma)), \gamma \in \Gamma)$$

has nonempty intersection. In particular $V((t^\Gamma))$ is complete for valuation-Cauchy nets. Thus $\mathcal{H}_\Gamma(H)$ is spherically complete even when its orthogonal projection theorem fails.

This is the three-duals report’s `duals:prop:spherical`; source 06 credits it there and both sources reprove it. In brief: a ball fixes all coefficients strictly below its radius exponent; nested balls impose compatible prescriptions; the vector x with the prescribed coefficients and zeros elsewhere has well-ordered support, because any subset S' of it and any $s \in S'$ give, below s , a subset of the well-ordered support of one ball’s centre; hence x lies in every ball. A Cauchy net has compatible stabilized truncations, and the same argument produces its limit. The argument uses no norm convergence in V . Spherical completeness here is for valuation balls only, not for arbitrary ordered-field norm balls.

The last sentence is informative: completeness here does not force ordinary completeness on coefficient subspaces. If $V \subset H$ is any ordinary linear subspace, even one that is not ordinarily closed, then $V((t^\Gamma))$ is valuation closed in $H((t^\Gamma))$: a coefficient outside V cannot be changed by a sufficiently small valuation perturbation. This is one source of the projection failure.

Proposition 3.4 (A first nonprojectable complete subspace [06]). *Let $H = \ell^2(\mathbb{N}, \mathbb{T})$ and $M = c_{00}((t^\Gamma)) \subseteq \mathcal{H}_\Gamma(H)$, where c_{00} is the space of finitely supported sequences. Then M is proper, valuation closed, and spherically complete, but $M^\perp = 0$. No point outside M has a nearest point in M .*

Proof. Closedness is the preceding paragraph and spherical completeness is [Theorem 3.3](#) with $V = c_{00}$. The constants e_n belong to M , so orthogonality to M forces every coordinate of every coefficient to vanish; thus $M^\perp = 0$. If $x \notin M$ and $m \in M$, then $r = x - m \neq 0$ has some $\langle r, e_n \rangle \neq 0$, and

$$\|x - (m + \langle r, e_n \rangle e_n)\|^2 = \|r\|^2 - |\langle r, e_n \rangle|^2 < \|r\|^2.$$

The correction belongs to M , excluding a nearest point. □

This subspace is *not* orthoclosed: $M^{\perp\perp} = \mathcal{H}_\Gamma(H)$. It is therefore not yet a witness to the failure of orthomodularity; [Theorem 8.7](#) supplies an orthoclosed one. The same phenomenon, a closed proper subspace with zero orthogonal complement, appears in the spectral report’s closed-range theorem `ihs:hh:thm:closedrange` and in the three-duals report’s closed hyperplane `duals:thm:hyperplane`.

3.3 Finite-dimensional Hilbert geometry survives

Proposition 3.5 ([03]). *Every finite-dimensional K -linear subspace of $\mathcal{H}_\Gamma(H)$ has an orthonormal basis and an orthogonal projection. If $u_1, \dots, u_d$ is such a basis, then*

$$P_M x = \sum_{j=1}^d \langle x, u_j \rangle u_j.$$

The coefficient-zero vectors $(u_j)_0$ form an ordinary orthonormal family, and the residue of M has ordinary dimension d .

Proof. Finite Gram–Schmidt uses only positive squared norms and their roots from [Proposition 3.1](#). The displayed formula is then a self-adjoint idempotent onto M . Since $\|u_j\| = 1$, one has $v(u_j) = 0$, and taking constant coefficients of the orthogonality identities gives ordinary orthonormality. For $x = \sum a_j u_j$, positivity gives $v(x) = \min_j v(a_j)$; there is no cancellation of its least coefficient because the leading coefficient vectors of the u_j are orthonormal. Thus integral vectors in M have precisely the residue span of those d vectors. $\square$

4 Coordinates, two summation levels, and duality

4.1 An exact coordinate-space description

Choose an ordinary orthonormal basis indexed by a set I , identifying H with $\ell^2(I, \mathbb{T})$. The coordinate map sends $x = \sum x_\gamma t^\gamma$ to $a_i = \sum_\gamma (x_\gamma)_i t^\gamma$.

Proposition 4.1 (Admissible surcomplex coordinate arrays [03, 06]). *The coordinate map identifies $\mathcal{H}_\Gamma(\ell^2(I))$ with the families $(a_i) \in K^I$ satisfying both conditions:*

$$\bigcup_{i \in I} \text{supp } a_i \text{ is well ordered,} \quad ([t^\gamma] a_i)_{i \in I} \in \ell^2(I) \quad \text{for every } \gamma.$$

The inner product is

$$[t^\delta] \langle x, y \rangle = \sum_{\alpha + \beta = \delta} \left(\sum_{i \in I} (x_\alpha)_i \overline{(y_\beta)_i} \right). \quad (4.1)$$

The outer sum is finite and each inner sum is ordinarily absolutely convergent.

Proof. The conditions are precisely that the coefficient rows define ordinary Hilbert vectors and that their nonzero support is a Hahn support; the support of the vector is exactly the displayed union. They provide the inverse coordinate map. Cauchy–Schwarz in the ordinary $\ell^2(I)$ proves absolute convergence of the inner sum; the support lemma proves finiteness of the outer sum. $\square$

Example 4.2 (Why a single formal coordinate sum is inadequate [03, 06]). The constant vector $x = (1/n)_{n \geq 1}$ belongs to $\mathcal{H}_\Gamma(\ell^2)$ and has its usual real squared norm $\sum_{n \geq 1} n^{-2}$. But the scalar family $(1/n^2)_{n \geq 1}$ is not strongly Hahn summable: infinitely many nonzero terms occur at exponent zero. Formula (4.1) uses ordinary Hilbert summation at a fixed coefficient, not that inadmissible formal sum. Conversely, $x = \sum_{n \geq 1} e_n t^{n/(n+1)}$ is a legitimate Hahn vector even though its monomial partial sums do not converge in valuation.

Thus “square summable surcomplex coordinates” must include a summation rule. Merely requiring finite partial squared sums to be bounded is far too weak: an infinite scalar already bounds every ordinary integer, so the constant family of coordinate values 1 would pass that test. The space is also generally larger than $K \otimes_\Gamma H$, which contains only series whose coefficient vectors lie in one finite-dimensional subspace; the exact completion distinctions are treated in the three-duals report, and no density of the algebraic scalar extension is assumed here.

4.2 What becomes of an orthonormal basis

The constant e_i remain orthonormal and separate all vectors by inner products. They need not be a valuation-topological Schauder basis. A nonzero ordinary tail, however small its ordinary Hilbert norm, has valuation zero. Its size therefore does not tend to zero at all surreal scales. Basis synthesis in this model is *layered*: ordinary Hilbert synthesis within each coefficient, then Hahn assembly across exponents. Parseval holds coefficientwise in the sense of (4.1), not as an unrestricted strong sum over i .

4.3 Riesz representation requires more than valuation continuity

The three-duals report [18] proves a detailed classification; the spectral report has its own counterexample (`ihs:hh:prop:riesz`). Two elementary obstructions are enough here. Both start from an everywhere-defined but ordinarily discontinuous linear functional on an infinite-dimensional H , obtained from a Hamel basis.

An infinitely bounded functional [03]. Let $\ell : H \rightarrow \mathbb{T}$ be such a functional and set $L(\sum x_\gamma t^\gamma) = \sum \ell(x_\gamma) t^\gamma$. It respects strong sums and satisfies $v(Lx) \geq v(x)$, hence is valuation continuous. It cannot equal $\langle x, y \rangle$ for a Hahn vector y : testing constants forces all nonconstant coefficients of y to vanish and represents ℓ by an ordinary Hilbert vector, a contradiction. For a fixed $\eta > 0$, this functional even satisfies $|Lx| \leq t^{-\eta} \|x\|$. The infinite bound dominates each individual ordinary coefficient ratio. Therefore the usual phrase “bounded functional” does not restore Riesz unless the meaning of boundedness is strengthened.

Proposition 4.3 (Even strict contractions need not be representable [06]). *If H is infinite dimensional and $q = t^\eta$ with $\eta > 0$, there is a K -linear strong functional $L : \mathcal{H}_\Gamma(H) \rightarrow K$ such that*

$$|Lx| < c \|x\| \quad (x \neq 0)$$

for every positive ordinary real c , but L is not representable.

Proof. Choose an orthonormal sequence (e_n) and a linear functional $f : H \rightarrow \mathbb{T}$ with $f(e_n) = n$, extended algebraically from its span by a Hamel basis. It is unbounded. Extend f coefficientwise to $\hat{f}$, and put $L = q\hat{f}$. Coefficientwise extension is strong, and $v(Lx) \geq v(x) + \eta > v(\|x\|)$, so the displayed inequality follows from leading exponents, independently of the size of the ordinary coefficient.

If $Lx = \langle x, y \rangle$, testing constants u and taking the coefficient at η would give $f(u) = \langle u, y_\eta \rangle_H$, a bounded functional, which is impossible. $\square$

By contrast, a globally Hahn-supported series of ordinary continuous functionals is represented, coefficient by coefficient, by a unique Hahn vector. Precisely: a K -linear functional is representable if and only if it is strong and each coefficient functional $u \mapsto [t^\gamma]Lu$ on the ordinary H is continuous. This restricted Riesz theorem is the three-duals report's `duals:thm:riesz` (the identification of the represented dual with $H'((t^\Gamma))$); source 06 reproves it (its Proposition 5.2) with the Baire support lemma [Lemma 5.1](#) below. Neither obstruction above is claimed as new.

4.4 Least field-valued norm bounds

Every ordinary bounded operator lifts to a constant Hahn operator ([Section 5](#)), and a norm bound means $C \in F_{\geq 0}$ with $\|\hat{A}x\| \leq C\|x\|$ for all x .

Theorem 4.4 (Least field-valued bound; repository theorem [06]). *Let $0 \neq A \in \mathcal{B}(H, G)$. Its constant lift $\hat{A}$ has a least nonnegative F -valued norm bound if and only if A attains its ordinary operator norm, and then the least bound is $\|A\|$. If $\|A\|$ is not attained, a nonnegative finite $C \in F$ is a bound exactly when $\text{st } C \geq \|A\|$, every positive infinite scalar is a bound, and the set of bounds has no least element.*

Proof. For $H = G$ this is the spectral report's norm-attainment criterion `ihs:hh:thm:norm`, whose proof describes all upper bounds exactly as stated. For the rectangular case `[merge]`, apply it to $T = \begin{pmatrix} 0 & 0 \\ A & 0 \end{pmatrix}$ on $H \oplus G$: $\|T\| = \|A\|$, T attains its norm exactly when A does, and $\|\hat{T}(x, y)\| = \|\hat{A}x\|$ with $\|x\| \leq \|(x, y)\|$, so $\hat{T}$ and $\hat{A}$ have the same bounds (test $y = 0$ for the converse). $\square$

The zero operator has least nonnegative bound zero and is excluded from the nonzero criterion. For example, $Ae_n = (1 - 1/(n+1))e_n$ is positive, self-adjoint and adjointable after lifting, but has no least F -valued operator-norm bound (the spectral report's `ihs:hh:ex:noleast`). Thus the order bound of [Corollary 5.3](#) is not a construction of an F -valued Banach C^* -algebra norm [06].

5 The intrinsic algebra of adjointable operators

5.1 Operators with a common Hahn support

For ordinary Hilbert spaces H, G , put

$$\mathcal{B}_\Gamma(H, G) = \mathcal{B}(H, G)((t^\Gamma)).$$

An element $A = \sum A_\gamma t^\gamma$ has one well-ordered support of nonzero ordinary bounded operators. It acts on all Hahn vectors by

$$(Ax)_\delta = \sum_{\gamma+\alpha=\delta} A_\gamma x_\alpha.$$

The support lemma makes these coefficient sums finite. Composition is Hahn convolution and $A^* = \sum A_\gamma^* t^\gamma$. There is no requirement that the numbers $\|A_\gamma\|$ be uniformly bounded as γ varies, and no growth condition. Write $v(A) = \min \text{supp } A$ for $A \neq 0$, and $v(0) = +\infty$. Then

$$v(Ax) \geq v(A) + v(x), \quad v(AB) \geq v(A) + v(B). \quad (5.1)$$

An operator is called *integral* if its valuation is nonnegative. An integral operator with an integral two-sided inverse preserves vector valuations exactly. Multiplication on either side by such an operator also preserves the valuation of any nonzero operator: apply (5.1) first to the operator and then to its inverse.

The apparent restriction to globally supported series is intrinsic to adjoints, rather than an optional technical convenience. The key step is a Baire lemma, isolated by source 06; it is Step 2 of the spectral report's proof of `ihh:hh:thm:adjoint`.

Lemma 5.1 (A Baire support lemma [06]). *Let $(A_\gamma)_{\gamma \in J}$ be bounded operators from an ordinary Hilbert space H to another, where J is a subset of an ordered set. If $\{\gamma : A_\gamma u \neq 0\}$ is well ordered for every $u \in H$, then $\{\gamma : A_\gamma \neq 0\}$ is well ordered.*

Proof. Otherwise dependent choice gives a strictly decreasing sequence $\gamma_1 > \gamma_2 > \dots$ of indices of nonzero operators. Each $H \setminus \ker A_{\gamma_n}$ is open and dense, because a proper closed subspace has empty interior. Baire's theorem gives a u in their intersection, contradicting the hypothesis on its support. $\square$

Theorem 5.2 (Automatic adjoint structure; repository input [03, 06]). *Let $A : \mathcal{H}_\Gamma(H) \rightarrow \mathcal{H}_\Gamma(G)$ and $B : \mathcal{H}_\Gamma(G) \rightarrow \mathcal{H}_\Gamma(H)$ be everywhere-defined K -linear maps satisfying*

$$\langle Ax, y \rangle = \langle x, By \rangle \quad (x \in \mathcal{H}_\Gamma(H), y \in \mathcal{H}_\Gamma(G)).$$

There is a unique member of $\mathcal{B}_\Gamma(H, G)$ producing the given map, and $B = A^$ coefficientwise. Conversely every member of $\mathcal{B}_\Gamma(H, G)$ is adjointable. In particular every everywhere-defined adjointable map is strong, valuation continuous, and order bounded. No continuity, strongness or common-support assumption is needed in the forward direction.*

Proof. The square case $H = G$ is `ihh:hh:thm:adjoint` of the spectral report, and both sources reprove it with the same three steps (bounded coefficients by the closed graph theorem, one well-ordered support by Lemma 5.1, identification on all vectors by testing against constants); they are not reprinted. The rectangular case reduces to the square one [merge]. On $\mathcal{H}_\Gamma(H \oplus G) = \mathcal{H}_\Gamma(H) \oplus \mathcal{H}_\Gamma(G)$ put $C(x, y) = (By, Ax)$. Then

$$\langle C(x, y), (x', y') \rangle = \langle By, x' \rangle + \langle Ax, y' \rangle = \langle y, Ax' \rangle + \langle x, By' \rangle = \langle (x, y), C(x', y') \rangle,$$

so C is an everywhere-defined self-adjoint map, and the square theorem gives $C = \sum_\gamma C_\gamma t^\gamma$ with $C_\gamma = C_\gamma^*$ bounded on $H \oplus G$ and well-ordered support. The lower-left blocks A_γ of the C_γ give the expansion of A , their upper-right blocks give that of B , and $C_\gamma = C_\gamma^*$ identifies the latter with A_γ^* . Uniqueness follows by evaluating on constants and extracting coefficients. The converse is coefficient comparison. Strongness, continuity and the order bound are `ihh:hh:cor:autobounded` applied to C , since $\|Ax\| = \|C(x, 0)\|$. For real H, G the same proof applies verbatim, as both sources observe. $\square$

Corollary 5.3 (Valuation and automatic boundedness [03, 06]). *For $0 \neq A \in \mathcal{B}_\Gamma(H, G)$,*

$$v(Ax) \geq v(A) + v(x), \quad v(A^*) = v(A), \quad v(A^*A) = 2v(A).$$

If $s = v(A)$ and c is an ordinary real with $c > \|A_s\|$, then $\|Ax\| \leq ct^s \|x\|$ for all x .

Proof. The first two assertions are the support bound and the coefficientwise adjoint. The leading coefficient of A^*A is $A_s^*A_s \neq 0$. The bound is `ihh:hh:cor:autobounded`: for nonzero x with leading term $t^\delta u$, the coefficient at $2(s + \delta)$ of $c^2 t^{2s} \|x\|^2 - \|Ax\|^2$ is $c^2 \|u\|^2 - \|A_s u\|^2 > 0$, also when $A_s u = 0$, since then the leading output moves to a higher exponent. $\square$

Remark 5.4. The theorem uses ordinary Hilbert completeness in its coefficient closed-graph and Baire arguments. Spherical completeness of the Hahn space alone does not supply those arguments for arbitrary coefficient vector spaces. Nor does the theorem classify all valuation-continuous maps: the three-duals report gives strictly larger classes.

5.2 Positive-order operator calculus

If $B \in \mathcal{B}_\Gamma^{>0}(H, H)$, then

$$(I + B)^{-1} = \sum_{n \geq 0} (-B)^n, \quad (I + B)^c = \sum_{n \geq 0} \binom{c}{n} B^n, \quad e^B = \sum_{n \geq 0} \frac{B^n}{n!}, \quad \log(I + B) = \sum_{n \geq 1} \frac{(-1)^{n+1} B^n}{n} \quad (5.2)$$

($c \in \mathbb{R}$) are legitimate strong Hahn operator series [06]. Their support lies in the additive monoid of $\text{supp } B$. At a fixed exponent there are finitely many words, so the usual one-variable formal identities survive substitution, even in the noncommutative coefficient algebra. No two noncommuting variables are ever substituted into an identity that presupposes commutativity. If $B = B^*$, then $I + B$ and $(I + B)^{1/2}$ are self-adjoint and positive definite, each I plus positive order; for x with leading coefficient u at exponent δ , the quadratic form of either operator has leading coefficient $\|u\|^2$ at exponent 2δ . This is *formal* smallness, not a real operator-norm estimate.

5.3 Finite-rank operators and self-adjoint inverses

For fixed Hahn vectors u, w , the map $x \mapsto \langle x, w \rangle u$ is adjointable, with adjoint $y \mapsto \langle y, u \rangle w$. Expanding its two vector supports gives a Hahn series of bounded finite-rank coefficient operators. Finite sums of these maps include every finite-dimensional orthogonal projection constructed in [Proposition 3.5](#).

If a self-adjoint everywhere-defined map B is bijective, then its algebraic inverse is self-adjoint: writing $x = Bu, y = Bv$ gives

$$\langle B^{-1}x, y \rangle = \langle u, Bv \rangle = \langle Bu, v \rangle = \langle x, B^{-1}y \rangle.$$

[Theorem 5.2](#) then puts B^{-1} in the Hahn operator algebra. This specific observation is used below, rather than an unproved general bounded-inverse theorem over K [03].

6 Orthogonal splitting and its residue normal form

6.1 Three equivalent formulations

Definition 6.1. A K -linear subspace $M \subseteq \mathcal{H}$ is *orthogonally split*, or *orthogonally complemented*, if $\mathcal{H} = M \oplus M^\perp$, where $M^\perp = \{x : \langle x, m \rangle = 0 \text{ for every } m \in M\}$. The word “split” below always means this orthogonal property. M is *orthoclosed* if $M = M^{\perp\perp}$.

Proposition 6.2 (Projection and best approximation [03]). *For a subspace $M \subseteq \mathcal{H}$, the following are equivalent:*

- (i) M is orthogonally split;
- (ii) M is the range of an everywhere-defined self-adjoint idempotent;
- (iii) every $x \in \mathcal{H}$ has a nearest point in M , with distances compared in the ordered field F .

The projection and all nearest points are unique. A split subspace is valuation closed, orthoclosed, and closed under all strong sums of its vectors.

Proof. A decomposition $x = m + n$ with $m \in M, n \in M^\perp$ is unique by positivity. Sending x to m defines a linear self-adjoint idempotent. Conversely the range and kernel of a self-adjoint idempotent are orthogonal, and its kernel equals the orthogonal complement of its range.

For such a decomposition and any $m' \in M$, Pythagoras gives

$$\|x - m'\|^2 = \|n\|^2 + \|m - m'\|^2.$$

Thus m is the unique nearest point. Conversely, let m be nearest to x and put $r = x - m$. If $n \in M$ and $\langle r, n \rangle \neq 0$, then $n \neq 0$ and the scalar $a = \langle r, n \rangle / \langle n, n \rangle$ gives

$$\|r - an\|^2 = \|r\|^2 - \frac{|\langle r, n \rangle|^2}{\|n\|^2} < \|r\|^2,$$

contradicting minimality. Hence $r \in M^\perp$ and the splitting follows. Finally $M = \ker(I - P)$ for its projection P ; automatic adjoint structure makes P continuous and strong. These two properties prove the two closure assertions, and $M^{\perp\perp} = \text{ran } P$ because $M^\perp = \text{ran}(I - P)$. $\square$

6.2 Every projection has ordinary residue

Lemma 6.3 (Integral projections [03, 06]). *Every projection P in $\mathcal{B}_\Gamma(H, H)$ is integral. If $P \neq 0$, then $v(P) = 0$. Its residue P_0 is an ordinary orthogonal projection, and $P - P_0$ has strictly positive order or is zero.*

Proof [03]. Let $s = v(P)$ when $P \neq 0$. Its nonzero leading coefficient P_s is ordinary self-adjoint, so $P_s^2 \neq 0$: otherwise $\|P_s u\|^2 = \langle P_s^2 u, u \rangle = 0$ for all u . Consequently $v(P^2) = 2s$. The identity $P^2 = P$ forces $2s = s$, hence $s = 0$. Taking residues gives $P_0^2 = P_0 = P_0^*$. $\square$

Second route to $v(P) \geq 0$ [06]. Pythagoras gives $\|Px\| \leq \|x\|$ for all x . If $s = v(P)$ were negative, a constant u with $P_s u \neq 0$ would give $v(Pu) = s < 0 = v(u)$, so $\|Pu\|$ would exceed $\|u\|$ by an infinite factor, contradicting this inequality. $\square$

For any subspace define its *integral residue space* by

$$\text{res } M = \{x_0 : x \in M, v(x) \geq 0\} \subseteq H. \quad (6.1)$$

This is an ordinary linear subspace, without a presumed ordinary closedness property.

Corollary 6.4 (Closed-residue obstruction [03, 06]). *If M is split with projection P , then*

$$\text{res } M = \text{ran } P_0.$$

In particular its residue space is ordinarily closed in H .

Proof. For integral $x = Px$ one has $x_0 = P_0x_0$. Conversely, for constant $u \in H$, the vector Pu is integral, belongs to M , and has residue P_0u . The range of an ordinary orthogonal projection is closed. $\square$

Ordinary closedness in this corollary is a substantially stronger restriction than valuation closedness of M . It detects the graph obstructions of Sections 8 and 9 without an appeal to spectral theory.

6.3 A global normal form using the classical direct rotation

Theorem 6.5 (Residue normal form [03, 06]). *A subspace $M \subseteq \mathcal{H}_\Gamma(H)$ is orthogonally split if and only if there are an ordinary closed subspace $M_0 \subseteq H$ and a Hahn unitary $U \in \mathcal{B}_\Gamma(H, H)$ satisfying $v(U - I) > 0$ or $U = I$, such that*

$$M = U\mathcal{H}_\Gamma(M_0). \quad (6.2)$$

Here $M_0 = \text{res } M$ is unique. Given the projection P onto M and P_0 onto M_0 , a canonical choice is

$$W = PP_0 + (I - P)(I - P_0) = I + (P - P_0)(2P_0 - I), \quad (6.3)$$

$$U = W(I - (P - P_0)^2)^{-1/2}, \quad (6.4)$$

the inverse square root being the formal binomial series about I . Then $UP_0 = PU$ and $P = UP_0U^*$. Hence every split subspace is unitarily isometric to the Hahn lift of an ordinary closed subspace, by a unitary whose residue is the identity.

Proof. Let M be split with projection P . By Theorem 5.2, $P \in \mathcal{B}_\Gamma(H, H)$, and by Lemma 6.3, P_0 is an ordinary orthogonal projection with $v(P - P_0) > 0$ (including $P = P_0$). These are exactly the hypotheses of the spectral report's support-controlled direct rotation `ihs:fr:lem:rotation`, with $P^\circ = P_0$: it gives $U^*U = UU^* = I$, $UP_0 = PU$ and $v(U - I) > 0$ for the unitary (6.4). Both sources reprove that lemma (by direct multiplication, $W^*W = WW^* = I - (P - P_0)^2$ and $WP_0 = PW$; the square $(P - P_0)^2$ commutes with both projections, so its binomial inverse square root commutes with W, W^*, P, P_0); the second expression for W in (6.3) is source 06's. It follows that $P = UP_0U^*$ and, since the constant projection P_0 acts coefficientwise, $\text{ran } P_0 = \mathcal{H}_\Gamma(M_0)$ with $M_0 = \text{ran } P_0$; so (6.2) holds. Corollary 6.4 gives $M_0 = \text{res } M$, proving uniqueness. Conversely, the operator UP_0U^* is an orthogonal projection onto the subspace in (6.2). $\square$

The formula (6.4) is the classical direct rotation of projections, the Kato–Nagy two-projection intertwiner; see Simon's accounts and attributions to earlier work [14, 15] and Kato [9]. The spectral report uses it for isolated finite spectral clusters. The point here is that automatic adjoints and integral projections place *every* split Hahn subspace in its domain of formal applicability. Neither the formula nor the rotation lemma is a new result; the unitary (6.4) is canonical as a formula but is not the only near-identity unitary conjugating P_0 to P .

6.4 Exact graph coordinates for a fixed residue

Corollary 6.6 (The positive-order graph chart [03, 06]). *Fix an ordinary closed $M_0 \subseteq H$ and decompose $H = M_0 \oplus M_0^\perp$. The split subspaces with residue M_0 are in bijection with*

$$X \in \mathcal{B}_\Gamma(M_0, M_0^\perp), \quad X = 0 \text{ or } v(X) > 0,$$

by $X \mapsto \text{Graph } X = \{(z, Xz) : z \in \mathcal{H}_\Gamma(M_0)\}$. For $J = (I, X)^\top$ the projection is

$$P_X = J(I + X^*X)^{-1}J^* = \begin{pmatrix} R_X & R_X X^* \\ X R_X & X R_X X^* \end{pmatrix}, \quad R_X = (I + X^*X)^{-1}. \quad (6.5)$$

The operator X is uniquely determined by the subspace.

Proof. Take the near-identity unitary U in [Theorem 6.5](#) and write its blocks relative to the displayed decomposition. The block $U_{00} = I + \text{positive-order terms}$ is invertible by [\(2.1\)](#). Its image of M_0 is therefore the graph of $X = U_{10}U_{00}^{-1}$, a positive-order operator with bounded coefficients. A graph determines its map uniquely.

Conversely $I + X^*X$ has a Neumann inverse of residue I . Formula [\(6.5\)](#) is self-adjoint and idempotent, since $J^*J = I + X^*X$; its range is exactly $\text{Graph } X$. Its residue is the projection onto the first summand. All products and inverses have support in the positive monoid generated by the supports of X, X^* . $\square$

One may read the bijection as an exact formal chart on the Grassmannian of split subspaces; no differential or manifold structure is asserted by that terminology [\[06\]](#). This is a classification of all split subspaces, not of all valuation-closed subspaces.

Proposition 6.7 (Closed residue is not sufficient [\[06\]](#), completed [\[merge\]](#)). *Let H be infinite dimensional, let $f : H \rightarrow \mathbb{T}$ be a discontinuous linear functional, let $\hat{f}$ be its coefficientwise extension, and put*

$$M = \{(x, q\hat{f}(x)) : x \in \mathcal{H}_\Gamma(H)\} \subseteq \mathcal{H}_\Gamma(H \oplus \mathbb{T}).$$

Then M is valuation closed and closed under strong sums, its residue $\text{res } M = H \oplus 0$ is closed, but $M^\perp = 0$ and $M \neq \mathcal{H}_\Gamma(H \oplus \mathbb{T})$. So M is neither split nor orthoclosed.

Proof. The map $\Phi(x, y) = y - q\hat{f}(x)$ is linear, strong and satisfies $v(\Phi(x, y)) \geq \min\{v(y), v(x) + \eta\}$, so it is valuation continuous and $M = \ker \Phi$ is closed and strong-sum closed. An integral $(x, q\hat{f}(x))$ has $v(x) \geq 0$, and its second component has valuation at least $v(x) + \eta > 0$; its residue is $(x_0, 0)$, and every $(u, 0)$, $u \in H$, is the residue of the constant choice $x = u$. So $\text{res } M = H \oplus 0$. If $(w, c) \perp M$, pairing with $(u, q\hat{f}(u))$ for constant u and extracting the coefficient at $\gamma + \eta$ gives

$$\langle u, w_{\gamma+\eta} \rangle_H + f(u) \overline{c_\gamma} = 0 \quad (u \in H).$$

If some $c_\gamma \neq 0$, then $f(u) = -\langle u, w_{\gamma+\eta} \rangle_H / \overline{c_\gamma}$ would be continuous; so $c = 0$, and then every coefficient of w is orthogonal to H , so $w = 0$. Finally $(0, 1) \notin M$. A split subspace with zero complement would be everything, and $M^{\perp\perp} = \mathcal{H}_\Gamma(H \oplus \mathbb{T}) \neq M$. $\square$

Source 06 remarks that “closed residue alone is insufficient”, pointing to coefficient maps such as those of [Proposition 4.3](#); the proposition supplies the details. It settles negatively the case left open by source 03, which makes no claim that an arbitrary valuation-closed subspace with closed residue automatically has a positive-order graph chart: this one has closed residue $H \oplus 0$ and no chart. The bounded-coefficient hypothesis in [Corollary 6.6](#) is part of the classification, not a disposable regularity condition. The subspace M is not orthoclosed; whether every *orthoclosed* subspace with closed residue splits is not decided here ([Question 15.2](#)).

7 Unitary structure and finite simultaneous straightening

Corollary 7.1 (Unitary orbit classification [06]). *Two projections on $\mathcal{H}_\Gamma(H)$ are unitarily conjugate if and only if their ordinary residue projections have the same range Hilbert dimension and the same kernel Hilbert dimension. These dimensions are cardinalities of ordinary orthonormal bases, not Hamel dimensions.*

Proof. A unitary and its inverse are contractions, so by the argument of the second route in Lemma 6.3 their supports are nonnegative. Taking residues in $U^*U = UU^* = I$ gives an ordinary unitary U_0 , and $UPU^* = P'$ gives $U_0P_0U_0^* = P'_0$, which preserves the two dimensions. Conversely, equality of the two dimension pairs gives an ordinary unitary between the residue decompositions. Compose its constant lift with the two straightening unitaries of Theorem 6.5. $\square$

Theorem 7.2 (A split unitary group and its infinitesimal logarithm [06]). *Every adjointable unitary on $\mathcal{H}_\Gamma(H)$ has a unique factorization*

$$U = \widehat{U}_0 \exp X, \quad U_0 \in \mathbf{U}(H), \quad X \in \mathcal{B}_\Gamma^{>0}(H, H), \quad X^* = -X.$$

Consequently residue gives a split exact sequence of groups

$$1 \longrightarrow \mathbf{U}_{1,\Gamma}(H) \longrightarrow \mathbf{U}(\mathcal{H}_\Gamma(H)) \xrightarrow{\text{res}} \mathbf{U}(H) \longrightarrow 1,$$

where $\mathbf{U}_{1,\Gamma}(H) = \{U \in \mathbf{U}(\mathcal{H}_\Gamma(H)) : U - I \in \mathcal{B}_\Gamma^{>0}(H, H)\}$. Exponential and logarithm are inverse bijections between the positive-order skew-adjoint series and $\mathbf{U}_{1,\Gamma}(H)$.

Proof. As in Corollary 7.1, U and U^* have nonnegative supports, and $U_0^*U_0 = U_0U_0^* = I$. Put $V = \widehat{U}_0^*U = I + Y$ with Y of positive order, and set $X = \log V$. Since V is unitary, formal one-variable identities give

$$X^* = \log V^* = \log V^{-1} = -\log V = -X.$$

Conversely $X^* = -X$ implies $(\exp X)^* \exp X = \exp(-X) \exp X = I$. Uniqueness follows from the residue and the formal inverse identities for exponential and logarithm. Constant lifting is a group-theoretic section. $\square$

The exponential parameter is not an additive group parameter: multiplication of noncommuting infinitesimal unitaries corresponds to the formal Baker–Campbell–Hausdorff law. No real norm convergence of that law is asserted. General formal exponential correspondences are established material [3]; the statement above specifies the exact unitary group in the present Hilbert setting.

Theorem 7.3 (Finite orthogonal decompositions straighten simultaneously [06]). *Let $P_1, \dots, P_r$ be pairwise orthogonal projections with $\sum_{j=1}^r P_j = I$, and put $Q_j = \widehat{(P_j)}_0$. Then a single near-identity adjointable unitary U satisfies*

$$UQ_jU^* = P_j \quad (1 \leq j \leq r).$$

An explicit choice is

$$W = \sum_{j=1}^r P_j Q_j, \quad U = W(W^*W)^{-1/2}.$$

The same conclusion holds for a finite orthogonal family not summing to I , after adjoining its complementary projection.

Proof. The Q_j form an ordinary orthogonal partition of I , and $W_0 = I$. Hence W is formally invertible and $W^*W = I + \text{positive order}$. Orthogonality gives $WQ_j = P_jQ_j = P_jW$. Taking adjoints, $Q_jW^* = W^*P_j$, so $Q_jW^*W = W^*P_jW = W^*WQ_j$: the operator W^*W commutes with each Q_j , and so does its inverse square root. Therefore $UQ_j = P_jU$. Moreover $U^*U = I$, and invertibility of U implies $UU^* = I$. Its residue is I . $\square$

For $r = 2$ this is [Theorem 6.5](#) with a different unitary. The finite qualification is essential: a countable family may fail to have even an orthogonal join, and countable simultaneous straightening can fail for two independent reasons ([Section 9.4](#)).

8 Amplified and infinitesimal graphs

8.1 The normal equation is the whole obstruction

Proposition 8.1 (General adjointable graph criterion [03]). *Let $A \in \mathcal{B}_\Gamma(H, G)$. Its graph is valuation closed and closed under strong Hahn sums. It is orthogonally split if and only if $I + A^*A$ is surjective on $\mathcal{H}_\Gamma(H)$. In that case this operator is invertible in $\mathcal{B}_\Gamma(H, H)$ and the graph projection is*

$$P = \begin{pmatrix} I \\ A \end{pmatrix} (I + A^*A)^{-1} \begin{pmatrix} I & A^* \end{pmatrix}.$$

Proof. The graph is the kernel of the continuous strong map $(x, y) \mapsto y - Ax$, which gives both closure properties. The operator $B = I + A^*A$ is injective because

$$\langle Bx, x \rangle = \|x\|^2 + \|Ax\|^2 > 0 \quad (x \neq 0).$$

If the graph is split, project $(f, 0)$ onto a point (x, Ax) . Orthogonality of the residual to every (u, Au) gives $Bx = f$. As f was arbitrary, B is onto. Conversely, if B is onto it is bijective and self-adjoint, so the observation at the end of [Section 5](#) gives B^{-1} in the adjointable algebra. Writing $Jx = (x, Ax)$, the identity $J^*J = B$ verifies that $JB^{-1}J^*$ is an orthogonal projection with range Graph A . $\square$

Positivity of $I + A^*A$ does not alone imply its surjectivity. This is exactly the step in an ordinary Hilbert-space graph argument that cannot be imported without further hypotheses.

Lemma 8.2 (Graphs of integral operators split [03, 06]). *If $B \in \mathcal{B}_\Gamma(V, W)$ is integral, then its graph is split in $\mathcal{H}_\Gamma(V \oplus W)$, and the residue of its projection is the ordinary projection onto Graph B_0 .*

Proof. The constant coefficient of $I + B^*B$ is $I + B_0^*B_0$, an ordinary bounded positive operator bounded below by I , hence boundedly invertible. Factoring out this constant reduces the inverse of $I + B^*B$ to a positive-order Neumann series, and [Proposition 8.1](#) applies. Taking residues in the projection formula gives the last assertion. $\square$

8.2 Finite and infinite scalar amplification of bounded operators

Theorem 8.3 (Exact amplification threshold [03]). *Let $T : H \rightarrow G$ be an ordinary bounded operator and $a \in K$. Then*

$$\text{Graph}(aT) \text{ is split} \iff a = 0 \text{ or } v(a) \geq 0 \text{ or } \text{ran } T \text{ is ordinarily closed.} \quad (8.1)$$

For nonzero infinite a , one has the exact residue formula

$$\text{res Graph}(aT) = \ker T \oplus \text{ran } T \subseteq H \oplus G. \quad (8.2)$$

When the graph is split, its projection has residue the ordinary projection onto $\text{Graph}(a_0T)$ in the finite case, and onto $\ker T \oplus \text{ran } T$ in the infinite case.

Proof. If $a = 0$ the graph is horizontal. If $v(a) \geq 0$, then aT is integral with residue a_0T , $a_0 = [t^0]a$, and Lemma 8.2 proves splitting and the finite-case residue assertion.

Now let $v(a) < 0$. Set $b = a^{-1}$ and $\eta' = v(b) > 0$, with $b_{\eta'} \neq 0$. A graph vector (x, y) is integral exactly when both components are integral. It satisfies $Tx = by$. The constant coefficient gives $Tx_0 = 0$. The coefficient at η' gives

$$Tx_{\eta'} = b_{\eta'}y_0,$$

because y has no negative exponents and b has least exponent η' . Hence $(x_0, y_0) \in \ker T \oplus \text{ran } T$. Conversely, for any $k \in \ker T, u \in H$,

$$x = k + bu, \quad y = Tu$$

is an integral graph vector with residue (k, Tu) . This proves (8.2). If the graph is split, Corollary 6.4 makes this residue space ordinarily closed, forcing $\text{ran } T$ to be closed in G .

For the converse assume $\text{ran } T$ closed. Write

$$H = N \oplus X, \quad N = \ker T, \quad G = Y \oplus Y^\perp, \quad Y = \text{ran } T.$$

The ordinary operator $T_1 = T|_X : X \rightarrow Y$ is a bounded isomorphism, by the ordinary bounded inverse theorem. Let $C_1 = T_1^*T_1$ on X ; it is boundedly invertible. On N , $B = I + \bar{a}aT^*T$ equals I . On X an inverse is

$$B_X^{-1} = (\bar{a}a)^{-1}C_1^{-1}(I + (\bar{a}a)^{-1}C_1^{-1})^{-1}. \quad (8.3)$$

The inner correction has positive valuation $2\eta'$, so its inverse is Hahn defined. This supplies a two-sided inverse to B and proves splitting. The residue projection is the ordinary projection onto (8.2), by Corollary 6.4. $\square$

Remark 8.4 (Geometric interpretation [03]). At a finite nonzero residue a_0 , the leading geometry is the ordinary sloped graph. If $v(a) > 0$, it is horizontal. At an infinite scale and with closed range, the nonkernel portion of the graph becomes vertical: its residue is $\ker T \oplus \text{ran } T$. A nonclosed ordinary range cannot be the residue of an orthogonal projection, so precisely this case obstructs the transition. The threshold depends on the sign of $v(a)$, not on a metric magnitude or a discrete value-group step. A failure of ordinary closed range becomes a failure of surreal best approximation after *any* infinite amplification, however small its negative valuation.

Example 8.5 (An explicit complete graph with no nearest point [03]). Let $H = G = \ell^2(\mathbb{N}_{\geq 1})$, $T = \text{diag}(1/n)$ and $A = q^{-1}T$. The range of T is not ordinarily closed: it contains all finite-support vectors but does not contain $h = (1/n)_{n \geq 1}$. Indeed $Tx = h$ would require $x = (1, 1, \dots) \notin \ell^2$. The graph of A is therefore not split.

There is also a direct witness to failure of best approximation. The vector $z = (h, 0)$ has no nearest point in the graph. Such a point (x, Ax) would solve the normal equation

$$(q^2I + T^2)x = q^2h. \quad (8.4)$$

Because T^2 is injective, the left side has valuation $v(x)$ for nonzero x . Thus $v(x) = 2\eta$. Its coefficient there would satisfy $T^2 x_{2\eta} = h$, requiring $x_{2\eta} = (n)_{n \geq 1}$, again not a Hilbert vector. The contradiction proves the assertion without a compact-operator spectral theorem.

This graph is valuation closed, strong-sum closed, and complete as a closed subspace of the complete valued space in [Theorem 3.3](#). These properties do not supply a nearest point. Source 03 asserts failure of an attained minimum here and needs no assertion about a distance infimum; for the flipped point $(0, h)$ the distances have no infimum at all ([Proposition 8.9](#)).

Corollary 8.6 (Enlarging the scale group cannot repair this graph [03]). *Let $\Gamma \hookrightarrow \Delta$ be an ordered group embedding, and extend coefficients and exponents to the Δ Hahn spaces. For ordinary T and $v(a) < 0$, the extended graph is split exactly when the original one is split, namely when $\text{ran } T$ is ordinarily closed. This holds also for noncofinal embeddings of value groups.*

Proof. The image of a still has negative valuation. [Theorem 8.3](#) applies over both groups with the same ordinary range condition. $\square$

8.3 Infinitesimal graphs of closed unbounded operators

Let H, G be ordinary Hilbert spaces and let $A : \text{Dom } A \subseteq H \rightarrow G$ be a closed densely defined ordinary operator. The domain is given its inherited Hilbert inner product, not its graph norm, when forming $\text{Dom } A((t^\Gamma))$. Write A_Γ for coefficientwise application on that space and put

$$M_q(A) = \{(x, qA_\Gamma x) : x \in \text{Dom } A((t^\Gamma))\} \subseteq \mathcal{H}_\Gamma(H \oplus G). \quad (8.5)$$

Although A can be unbounded in ordinary norm, A_Γ does not lower valuation on its domain. Thus the graph is infinitesimally close to the horizontal domain in a valuation sense. It is not the graph of an everywhere-defined bounded-coefficient Hahn operator unless A is bounded.

Theorem 8.7 (Exact infinitesimal-graph criterion [06]). *For every closed densely defined A and every $\eta \in \Gamma_{>0}$, with $q = t^\eta$:*

- (i) $M_q(A)$ is a strong K -linear subspace, valuation closed and spherically complete.
- (ii) Its orthogonal complement is

$$M_q(A)^\perp = \{(-qA_\Gamma^* y, y) : y \in \text{Dom } A^*((t^\Gamma))\}, \quad (8.6)$$

where A_Γ^* means coefficientwise application of the ordinary unbounded adjoint A^* . In particular $M_q(A)$ is orthoclosed.

- (iii) Its residue is exactly $\text{res } M_q(A) = \text{Dom } A \oplus \{0\}$.
- (iv) It is orthogonally split if and only if $\text{Dom } A = H$, equivalently if and only if the closed operator A is bounded.

Proof. Strong linearity and completeness. Coefficientwise application of any linear map preserves strong sums on its domain: supports only shrink, and each coefficient sum is finite. Scalar convolution is likewise compatible. For $x \neq 0$,

$$v((x, qA_\Gamma x)) = v(x), \quad (8.7)$$

because the second component has valuation at least $v(x) + \eta$. The graph parametrization is therefore a valuation isometry from $\text{Dom } A((t^\Gamma))$, which is spherically complete by [Theorem 3.3](#), even though $\text{Dom } A$ is usually incomplete in its inherited ordinary norm.

Orthogonality. If (u, w) is orthogonal to the graph, testing a constant $k \in \text{Dom } A$ and extracting exponent γ gives

$$\langle k, u_\gamma \rangle_H + \langle Ak, w_{\gamma-\eta} \rangle_G = 0 \quad (k \in \text{Dom } A).$$

By the definition of the ordinary densely defined adjoint, this says $w_{\gamma-\eta} \in \text{Dom } A^*$ and $A^*w_{\gamma-\eta} = -u_\gamma$. Varying γ proves (8.6); the reverse inclusion follows from finite Hahn coefficient regrouping. Taking a second orthogonal complement, by the same computation with $-A^*$ in place of A and the coordinates exchanged, replaces A by A^{**} . The ordinary theorem $A^{**} = A$ for closed densely defined operators [9] gives orthoclosedness. Orthogonal complements are valuation closed because pairings with a fixed vector are continuous; this proves the remaining closedness assertion in (i).

Residue and complementation. By (8.7), an integral graph vector has $v(x) \geq 0$ and second component of positive valuation. Its residue is $(x_0, 0)$ with $x_0 \in \text{Dom } A$. Every such residue is realized by a constant x . If the graph were split, Corollary 6.4 would make $\text{Dom } A$ closed in H . It is dense, so $\text{Dom } A = H$, and the ordinary closed graph theorem makes A bounded. Conversely, if A is bounded, $q\hat{A}$ has positive order and Corollary 6.6 gives its orthogonal graph projection. $\square$

Corollary 8.8 (Explicit absence of best approximation [06]). *If A is unbounded and $h' \in H \setminus \text{Dom } A$, the constant vector $(h', 0)$ has no nearest point in $M_q(A)$.*

Proof. By the proof of Proposition 6.2, a nearest point m in a linear subspace must leave an orthogonal residual. Suppose $(h', 0) = m + n$ with $m \in M_q(A)$ and $n \in M_q(A)^\perp$. Pythagoras makes both vectors integral. By Theorem 8.7(iii) and (8.6), their residues lie in $\text{Dom } A \oplus 0$ and $0 \oplus \text{Dom } A^*$ respectively. Hence $h' \in \text{Dom } A$, a contradiction. $\square$

The theorem is an explicit mechanism behind the abstract Solèr obstruction (Section 15.2): infinitesimal scaling preserves the ordinary domain defect in the residue, while valuation completeness ignores the ordinary metric incompleteness of that domain.

Proposition 8.9 (Exact lower cut of the distances [06]). *Let $H = G = \ell^2(\mathbb{N}_{\geq 1})$, $\Lambda = \text{diag}(n)$ with $\text{Dom } \Lambda = \{x : \sum n^2 |x_n|^2 < \infty\} = \text{ran } T$, and $z = (h, 0)$ with $h = (1/n)$. The nonnegative lower bounds of $\{\|z - m\| : m \in M_q(\Lambda)\}$ are precisely zero and the positive infinitesimals of F . In particular, no distance infimum exists in F .*

Proof. Λ is closed, self-adjoint and unbounded ($\|\Lambda e_n\| = n$), and $h \notin \text{Dom } \Lambda$. If a graph vector m has negative valuation, then $z - m$ has negative valuation and infinite norm. If it is integral, its constant coefficient is $(x_0, 0)$ with $x_0 \in \text{Dom } \Lambda$, so $h - x_0 \neq 0$, and $\|z - m\|$ has positive ordinary standard part. Thus every positive infinitesimal is a lower bound. Conversely let $h^{(N)}$ be the finite truncation of h . For $m_N = (h^{(N)}, q\Lambda h^{(N)})$,

$$\text{st}\|z - m_N\| = \|h - h^{(N)}\|_H \longrightarrow 0$$

in the ordinary real sense. No positive bound with positive standard part can be a lower bound, and no infinite positive bound either. The set of zero and the positive infinitesimals has no greatest member, since doubling a positive infinitesimal gives a larger one and q is one. Therefore the distance set has no infimum. $\square$

The cut is exactly the one of the three-duals report's `duals:thm:distance`, proved there for the closed hyperplane M_Γ of `duals:thm:hyperplane`, with the same argument (ordinary density of the residue plus valuation zero of every integral difference). Source 06 does not cite it. The new point of Proposition 8.9 is only that the subspace is now orthoclosed; the hyperplane has zero orthogonal complement and is not.

8.4 The common criterion: scaled closed linear relations

The two graph theorems are related, but neither implies the other. [Theorem 8.3](#) treats bounded T with arbitrary kernel and range, and also finite scalars; [Theorem 8.7](#) treats unbounded A and also computes the orthogonal complement. After exchanging the coordinates, both are graphs of a closed linear relation multiplied by a positive infinitesimal in the second coordinate.

A closed linear relation $R \subseteq H \oplus G$ is a closed linear subspace. Its domain and multivalued part are

$$\text{dom } R = \{x : (x, y) \in R \text{ for some } y\}, \quad \text{mul } R = \{y : (0, y) \in R\};$$

$\text{mul } R$ is closed, while $\text{dom } R$ need not be. Coefficientwise, $R((t^\Gamma)) \subseteq \mathcal{H}_\Gamma(H \oplus G)$ consists of the Hahn vectors all of whose coefficients lie in R . For $b \in K$ with $v(b) > 0$ put

$$M_b(R) = \{(x, by) : (x, y) \in R((t^\Gamma))\} \subseteq \mathcal{H}_\Gamma(H \oplus G).$$

Theorem 8.10 (Scaled closed relations [merge]). *Let $R \subseteq H \oplus G$ be a closed linear relation and $b \in K$ with $v(b) > 0$. Then*

$$\text{res } M_b(R) = \text{dom } R \oplus \text{mul } R, \tag{8.8}$$

and the following are equivalent: $M_b(R)$ is split; $\text{res } M_b(R)$ is ordinarily closed; $\text{dom } R$ is ordinarily closed. When they hold,

$$M_b(R) = \text{Graph}(b\widehat{A_R}) \oplus (0 \oplus \mathcal{H}_\Gamma(\text{mul } R)), \tag{8.9}$$

an orthogonal sum, where $A_R : \text{dom } R \rightarrow (\text{mul } R)^\perp$ is the bounded operator part of R .

Proof. Residue. Let $(x, by) \in M_b(R)$ be integral, with $(x, y) \in R((t^\Gamma))$. Then $v(x) \geq 0$ and $(x_0, y_0) \in R$, so $x_0 \in \text{dom } R$. The residue of the second component is $[t^0](by) = \sum_{\alpha+\beta=0} b_\alpha y_\beta$ with $\alpha \in \text{supp } b \subseteq \Gamma_{>0}$, so only coefficients y_β with $\beta < 0$ occur. For $\beta < 0$, $x_\beta = 0$, so $(0, y_\beta) \in R$ and $y_\beta \in \text{mul } R$. Hence $\text{res } M_b(R) \subseteq \text{dom } R \oplus \text{mul } R$. Conversely, for $u \in \text{dom } R$ choose $(u, y) \in R$; the constant pair gives the integral vector (u, by) with residue $(u, 0)$. For $m \in \text{mul } R$, the vector $(0, b^{-1}m)$ lies in $R((t^\Gamma))$, since each of its coefficients is a multiple of $(0, m)$, and gives $(0, m) \in M_b(R)$ with residue $(0, m)$. The residue space is linear, so it contains $\text{dom } R \oplus \text{mul } R$.

Necessity. If $M_b(R)$ is split, [Corollary 6.4](#) makes $\text{dom } R \oplus \text{mul } R$ closed in $H \oplus G$, and then $\text{dom } R = \{u : (u, 0) \in \text{dom } R \oplus \text{mul } R\}$ is closed in H . Conversely, $\text{mul } R$ is always closed, so $\text{dom } R \oplus \text{mul } R$ is closed whenever $\text{dom } R$ is.

Sufficiency. Let $H_1 = \text{dom } R$ be closed and $G_1 = (\text{mul } R)^\perp$. For $u \in H_1$ the set $\{y : (u, y) \in R\}$ is a coset of $\text{mul } R$; let $A_R u$ be its unique element in G_1 . Then $A_R : H_1 \rightarrow G_1$ is linear, and its graph is $R \cap (H \oplus G_1)$, which is closed. As H_1 is a Hilbert space, the closed graph theorem makes A_R bounded. Every $(u, y) \in R$ is $(u, A_R u) + (0, y - A_R u)$ with $y - A_R u \in \text{mul } R$, and this decomposition is given by bounded ordinary projections, so it holds coefficientwise: $R((t^\Gamma)) = \{(u, \widehat{A_R u})\} + (0 \oplus \text{mul } R((t^\Gamma)))$. Multiplying the second coordinates by b , and using that b is invertible, gives (8.9). The first summand is the graph of the positive-order operator $b\widehat{A_R} \in \mathcal{B}_\Gamma(H_1, G_1)$, which is split in $\mathcal{H}_\Gamma(H_1 \oplus G_1)$ by [Corollary 6.6](#); extending its projection by zero on $\mathcal{H}_\Gamma((H_1 \oplus G_1)^\perp)$ gives a projection on $\mathcal{H}_\Gamma(H \oplus G)$. The second summand is the constant split subspace $\mathcal{H}_\Gamma(0 \oplus \text{mul } R)$, which is orthogonal to $\mathcal{H}_\Gamma(H \oplus G_1)$. The sum of two projections with orthogonal ranges is the projection onto the sum ([Proposition 9.1](#)), so $M_b(R)$ is split. $\square$

Corollary 8.11 (Both graph theorems [merge]). (a) *For bounded $T : H \rightarrow G$ and $v(a) < 0$, the exchange of coordinates $(x, y) \mapsto (y, x)$ maps $\text{Graph}(aT)$ onto $M_{a^{-1}}(R)$ with $R = \{(Tx, x) : x \in$*

$H\} \subseteq G \oplus H$, for which $\text{dom } R = \text{ran } T$ and $\text{mul } R = \ker T$. *Theorem 8.10* gives the residue (8.2) and the infinite case of *Theorem 8.3*.

- (b) For closed densely defined A , $M_q(A) = M_q(\text{Graph } A)$ with $\text{dom} = \text{Dom } A$ and $\text{mul} = 0$; *Theorem 8.10* gives *Theorem 8.7(iii)* and (iv).

The two cases meet exactly when T is bounded and injective with dense range and $A = T^{-1}$: then $\text{Graph}(q^{-1}T)$ and $M_q(T^{-1})$ are exchanged by the coordinate flip. In particular, for $T = \text{diag}(1/n)$ the subspace of *Example 8.5* is the flip of $M_q(\Lambda)$.

Proof. For (a), $\text{Graph}(aT)$ flipped is $\{(aTx, x)\}$; substituting $x = a^{-1}x'$ gives $\{(Tx', a^{-1}x')\} = M_{a^{-1}}(R)$, because $R((t^\Gamma)) = \{(Tx', x') : x' \in \mathcal{H}_\Gamma(H)\}$. The flip is a constant unitary, so it preserves splitting and maps residues by the ordinary flip. For (b), the coefficients of $(x, A_\Gamma x)$ lie in $\text{Graph } A$ exactly when those of x lie in $\text{Dom } A$. The last assertion is (a) with $\ker T = 0$ and $\{(Tx, x)\} = \text{Graph}(T^{-1})$. $\square$

The relation theorem does not replace the two source theorems: it gives neither the finite-scalar clause of *Theorem 8.3* nor the orthogonal complement and orthoclosedness in *Theorem 8.7(ii)*.

9 The projection poset: meets, kernels and countable joins

9.1 What survives: orthocomplements and orthogonal sums

Let $\text{Spl}(\mathcal{H})$ be the collection of split K -linear subspaces, ordered by inclusion. Its bottom and top are 0 and $\mathcal{H}$. Sending a projection to its range is an order isomorphism from the poset of projections, ordered by $P \leq Q \iff PQ = QP = P$, onto $\text{Spl}(\mathcal{H})$ (source 06 works with that poset of projections). Orthocomplementation, $P \mapsto I - P$, is an order-reversing involution, by the defining orthogonal decompositions.

Proposition 9.1 (Orthomodular poset structure [03]). *If $M, N \in \text{Spl}(\mathcal{H})$ and $M \perp N$, then $M \oplus N$ is split and is their least upper bound in $\text{Spl}(\mathcal{H})$. If $M \subseteq N$, then $N \cap M^\perp$ is split and*

$$N = M \oplus (N \cap M^\perp).$$

Consequently $\text{Spl}(\mathcal{H})$ is an orthomodular poset.

Proof. Let P_M, P_N be the unique projections. Orthogonality gives $P_M P_N = P_N P_M = 0$, so $P_M + P_N$ is a self-adjoint idempotent with range $M \oplus N$. Every subspace containing both contains their sum, which proves the least-upper-bound assertion.

If $M \subseteq N$, then $P_N P_M = P_M$ and, taking adjoints, $P_M P_N = P_M$. The operator $P_N - P_M$ is a self-adjoint idempotent. Its range is exactly $N \cap M^\perp$, proving both splitting and the displayed orthomodular identity. These properties, together with orthocomplementation and the bounds 0, $\mathcal{H}$, are the asserted orthomodular-poset axioms. They do not assert that arbitrary binary meets or joins exist. $\square$

By induction, finite mutually orthogonal families have joins, their sums. *Theorem 9.5* shows that countable ones need not.

9.2 Two good subspaces with a bad intersection

Theorem 9.2 (Exact finite-dimensional/lattice dichotomy [03]; one direction also [06]). *Let $\Gamma \neq 0$ be any set-sized ordered abelian group. For an ordinary real or complex Hilbert space H ,*

$$\text{Spl}(\mathcal{H}_\Gamma(H)) \text{ is a lattice} \iff \dim H < \infty.$$

If H is infinite dimensional, there are two split subspaces whose intersection is not split and which have no greatest lower bound in $\text{Spl}(\mathcal{H}_\Gamma(H))$. Their orthogonal complements have no least upper bound.

Proof. First work in $\mathcal{H}_\Gamma(H_1 \oplus H_1 \oplus H_1)$ with $H_1 = \ell^2(\mathbb{N}_{\geq 1})$. With $T = \text{diag}(1/n)$ and $q = t^\eta$, define

$$\begin{aligned} M &= \{(x, y, 0) : x, y \in \mathcal{H}_\Gamma(H_1)\}, \\ N &= \{(x, y, Tx - qy) : x, y \in \mathcal{H}_\Gamma(H_1)\}. \end{aligned}$$

The first is a constant orthogonal summand. The second is the graph of the integral adjointable operator

$$B(x, y) = Tx - qy, \tag{9.1}$$

which is split by Lemma 8.2. Their intersection is

$$M \cap N = \ker B \oplus 0 = \{(x, q^{-1}Tx, 0) : x \in \mathcal{H}_\Gamma(H_1)\}. \tag{9.2}$$

Its residue space is $0 \oplus \text{ran } T \oplus 0$, by (8.2) and $\ker T = 0$. That space is not ordinarily closed. Corollary 6.4 therefore proves that the intersection is not split.

Suppose a greatest lower bound L of M, N existed in the split poset. For every nonzero $z \in M \cap N$, the one-dimensional space Kz is split by finite-dimensional orthogonal projection, and is a common lower bound. Greatestness implies $Kz \subseteq L$ for all such z . It follows that $M \cap N \subseteq L$, whereas the lower-bound condition gives $L \subseteq M \cap N$. This would make the nonsplit intersection itself split, a contradiction. An order-reversing involution exchanges meets and joins, so $M^\perp, N^\perp$ have no least upper bound.

Every infinite-dimensional Hilbert space contains a closed isometric copy of $H_1 \oplus H_1 \oplus H_1$, obtained by partitioning a countable orthonormal family into three infinite families and taking their closed spans. Extend the two projections by zero on the ordinary orthogonal complement. They give split subspaces in $\mathcal{H}_\Gamma(H)$ with the same intersection. Its residue is still nonclosed in H , and the preceding no-meet argument applies unchanged. All choices here are set-sized.

Finally, if $\dim H = d < \infty$, then $\mathcal{H}_\Gamma(H)$ is isometric to K^d (or F^d in the real case). Every linear subspace has a finite orthonormal basis by Proposition 3.5, hence is split. Ordinary algebraic intersection and sum are meet and join in the full subspace lattice, so the split poset is a lattice. $\square$

Source 06 proves the direction “not a lattice” for ℓ^2 (its Theorem 9.3) with the same construction: its row operator $(x, y) \mapsto \tau x - Sy$ is (9.1) with the two input coordinates exchanged, and it transports the example to ℓ^2 by an ordinary unitary $\ell^2 \oplus \ell^2 \oplus \ell^2 \cong \ell^2$. The theorem does not conflict with Theorem 6.5: that theorem classifies each split subspace individually, and intersections need not remain in the classified family; taking residues does not preserve intersections, even when both subspaces split.

Remark 9.3 (Relation to existing non-Archimedean obstructions [03]). The general phenomenon of nonorthomodularity in other non-Archimedean Hilbert-like models is not new; Aguayo–Nova

give such results for c_0 -type spaces [1], and Solèr’s theorem excludes orthomodularity over these fields altogether (Section 15.2). Here the coefficient-Hilbert model, the complete classification of split subspaces, and the precise criterion “finite ordinary coefficient dimension” identify a particular obstruction. In particular, the poset of *split* subspaces remains orthomodular, but ceases to be a lattice. This is more specific than saying that some closed subspaces lack orthogonal complements.

At the level of inner-product geometry, $\Gamma = 0$ gives the ordinary Hilbert space and its complete closed-subspace lattice even in infinite dimension. The earlier comparison of valuation and norm topologies assumed $\Gamma \neq 0$ and is not being applied to this degenerate case. Thus nontrivial scale structure is an essential hypothesis of the dichotomy, not just a notational preference.

9.3 A positive operator with a noncomplemented kernel

Theorem 9.4 (An adjointable positive operator with a noncomplemented kernel [06]). *For every nonzero Γ , the adjointable positive self-adjoint operator on $\mathcal{H}_\Gamma(\ell^2 \oplus \ell^2)$*

$$L = B^*B = \begin{pmatrix} T^2 & -qT \\ -qT & q^2I \end{pmatrix}, \quad (9.3)$$

with B as in (9.1), has a valuation-closed, orthoclosed, spherically complete kernel that is not orthogonally split. It is order bounded and has only the three exponents $0, \eta, 2\eta$.

Proof. All coefficients in (9.3) are ordinary bounded operators, so the Hahn operator is adjointable and order bounded. Positivity follows from $\langle Lz, z \rangle = \langle Bz, Bz \rangle$. This identity and anisotropy give $\ker L = \ker B = \{(x, q^{-1}Tx)\} = \text{Graph}(q^{-1}T)$, which by Corollary 8.11 is the flip of $M_q(\Lambda)$. The flip is a constant unitary, so Theorem 8.7 applies. $\square$

Thus this obstruction is already polynomial in a single infinitesimal; it is not caused by a divergent formal inverse or a complicated value group. It also shows that the adjointable operator algebra is not a Rickart $*$ -algebra: not every element has a projection-generated right annihilator [06]. Put $H = \ell^2 \oplus \ell^2$. If the right annihilator $\{X \in \mathcal{B}_\Gamma(H, H) : LX = 0\}$ of L were $\{PY : Y \in \mathcal{B}_\Gamma(H, H)\}$ for a projection P , then $LP = 0$ would imply $\text{ran } P \subseteq \ker L$. For each $x \in \ker L$, the rank-one operator $z \mapsto \langle z, e \rangle x$, for a fixed unit vector e , belongs to that annihilator and would force $x \in \text{ran } P$. Thus $\text{ran } P = \ker L$, a contradiction.

9.4 A mutually orthogonal countable family with no join

Let (e_n) and (f_n) be the standard orthonormal bases of the two summands of $\ell^2 \oplus \ell^2$. Define

$$v_n = e_n + nqf_n, \quad w_n = -nqe_n + f_n,$$

and let P_n and P'_n be the rank-one projections onto Kv_n and Kw_n . In the n th two-dimensional block,

$$P_n = \frac{1}{1 + n^2q^2} \begin{pmatrix} 1 & nq \\ nq & n^2q^2 \end{pmatrix}, \quad P'_n = \frac{1}{1 + n^2q^2} \begin{pmatrix} n^2q^2 & -nq \\ -nq & 1 \end{pmatrix}. \quad (9.4)$$

They vanish outside that block. The P_n are mutually orthogonal, and $P_m P'_n = 0$ for all m, n .

Theorem 9.5 (No countable orthogonal join at one scale [06]). *For every nonzero Γ , the family $(P_n)_{n \geq 1}$ in (9.4) has no least upper bound among the projections on $\mathcal{H}_\Gamma(\ell^2 \oplus \ell^2)$, although I is an upper bound. Every individual projection has support in the same well-ordered monoid $\mathbb{N}\eta$. Thus common well-ordered scalar support does not restore countable projection completeness.*

Proof. The coordinate relations $\langle (x, y), w_n \rangle = 0$ for all n say precisely $y_n = nqx_n$ for the n th coordinates. Reconstructing Hilbert coefficients coefficient by coefficient, as in [Corollary 8.11](#), gives

$$\bigcap_{n \geq 1} \ker P'_n = M_q(\Lambda).$$

Suppose Q were the least upper bound of all P_n . Since $I - P'_n$ is an upper bound for every n , minimality gives $Q \leq I - P'_n$ for every n , so $\text{ran } Q \subseteq M_q(\Lambda)$. On the other hand, $v_n \in \text{ran } Q$ for all n , and $\text{res } v_n = (e_n, 0)$. By [Corollary 6.4](#), the ordinary subspace $\text{res } \text{ran } Q$ is closed. It contains every $(e_n, 0)$, hence all of $\ell^2 \oplus 0$. But it must lie in $\text{res } M_q(\Lambda) = \text{Dom } \Lambda \oplus 0$, which is proper. This is impossible. The support assertion follows by expanding $(1 + n^2q^2)^{-1}$. $\square$

Corollary 9.6 (Finite straightening has no countable analogue [06]). *There is no adjointable unitary simultaneously sending the constant rank-one projections Q_n onto Ke_n to all the P_n in (9.4).*

Proof. If such a unitary U existed, it would have nonnegative support ([Corollary 7.1](#)), and its residue would preserve each constant line ∇e_n , since $(P_n)_0 = Q_n$. Write $U = \sum_{\gamma \geq 0} U_\gamma t^\gamma$. The coefficient at η of UQ_nU^* is $\sum_{\alpha+\beta=\eta} U_\alpha Q_n U_\beta^*$ with $\alpha, \beta \in \text{supp } U$; the fixed well-ordered support of U allows only finitely many such pairs, including pairs of two strictly positive exponents summing to η , and their operator norms are bounded independently of n , since $\|Q_n\| = 1$. Thus $\sup_n \|[t^\eta]UQ_nU^*\| < \infty$. Yet (9.4) gives P_n an exponent- η coefficient of norm n . This contradiction proves the claim. $\square$

The proof retains possible intermediate exponents of U ; discarding them would leave a gap at a nondiscrete value group. The obstruction here is *ordinary boundedness of coefficient operators*, not support shape.

Proposition 9.7 (Incoherent supports prevent simultaneous conjugation [06]). *Suppose Γ contains $\mathbb{Q}\eta$. Let $\tilde{P}_n$ project onto $K(e_n + t^{\eta/n}f_n)$. Each $\tilde{P}_n$ is a legitimate projection, but no adjointable unitary conjugates the constant projections Q_n onto Ke_n simultaneously to the $\tilde{P}_n$.*

Proof. If $S = \text{supp } U$ is the common well-ordered operator support, then $\text{supp}(UQ_nU^*) \subseteq S + S$ for every n . The set $S + S$ is well ordered, so it cannot contain the strictly decreasing sequence $\eta, \eta/2, \eta/3, \dots$, whereas the off-diagonal coefficient of $\tilde{P}_n$ at η/n is nonzero. $\square$

Thus infinite orthogonal synthesis has two genuinely distinct requirements: a coherent common support and bounded ordinary coefficient synthesis. Finite families automatically meet both; arbitrary countable families do not. These are necessary conditions only; no sufficient condition is proved ([Question 15.3](#)).

10 Metric rigidity and the exact positivity threshold

The preceding classification concerns the canonical Hahn extension of an ordinary Hilbert inner product. There are other positive definite forms on the same vector space. Even valuation completeness and a positive surreal coercivity bound do not force such a form to be isometrically standard. The relevant threshold is an *ordinary real* lower bound on its residue operator [06].

Let $H \neq 0$, and let

$$\Theta = \widehat{\Theta}_0 + \Theta_+ \in \mathcal{B}_\Gamma(H, H), \quad \Theta = \Theta^*, \quad v(\Theta_+) > 0 \text{ or } \Theta_+ = 0, \quad (10.1)$$

where $\Theta_0 \in \mathcal{B}(H)$ is ordinary self-adjoint and strictly positive in the pointwise sense $\langle \Theta_0 u, u \rangle_H > 0$ for $u \neq 0$. Strict positivity here does not mean uniform positivity. The form

$$\langle x, y \rangle_\Theta = \langle \Theta x, y \rangle \quad (10.2)$$

is positive definite: if x_δ is the first nonzero coefficient of x , its squared length has first term $t^{2\delta} \langle \Theta_0 x_\delta, x_\delta \rangle_H$. Its positive square root exists by the same binomial construction as in [Proposition 3.1](#), and

$$v(\|x\|_\Theta) = v(x) = v(\|x\|). \quad (10.3)$$

In particular the valued-space completeness properties of [Theorem 3.3](#) are unchanged. This concerns valuation balls; it does not upgrade the assertion to arbitrary ordered-field norm balls.

Theorem 10.1 (Exact isometric rigidity threshold [06]). *For Θ as in (10.1), the following are equivalent:*

- (i) *There is a K -linear bijection $J : \mathcal{H}_\Gamma(H) \rightarrow \mathcal{H}_\Gamma(H)$ with*

$$\langle Jx, Jy \rangle = \langle x, y \rangle_\Theta \quad (x, y \in \mathcal{H}_\Gamma(H)).$$

No continuity, strongness, or adjointability of J is assumed.

- (ii) *The residue Θ_0 is uniformly positive: for some ordinary real $c > 0$, $\langle \Theta_0 u, u \rangle_H \geq c \|u\|_H^2$ for every $u \in H$.*

When (ii) holds, an isometry is explicitly

$$J = (I + \Theta')^{1/2} \widehat{\Theta_0^{1/2}}, \quad \Theta' = \widehat{\Theta_0^{-1/2}} \Theta_+ \widehat{\Theta_0^{-1/2}}. \quad (10.4)$$

Proof. Assume (ii). Ordinary positive operator theory gives bounded $\Theta_0^{1/2}$ and $\Theta_0^{-1/2}$. The self-adjoint Θ' has positive order, so its formal binomial square root and inverse square root exist. The operator in (10.4) is bijective and satisfies

$$J^* J = \widehat{\Theta_0^{1/2}} (I + \Theta') \widehat{\Theta_0^{1/2}} = \Theta.$$

This proves (i), without a commutation assumption on Θ_0 and Θ_+ .

Conversely suppose (i). Since J is onto, for every z we have

$$\langle Jx, z \rangle = \langle \Theta x, J^{-1}z \rangle = \langle x, \Theta J^{-1}z \rangle.$$

Thus J has an everywhere-defined adjoint $J^* = \Theta J^{-1}$, and [Theorem 5.2](#) applies: J is a Hahn series of ordinary bounded operators. Since $J^* J = \Theta$ and $v(\Theta) = 0$, [Corollary 5.3](#) gives $2v(J) = 0$, hence $v(J) = 0$. Its residue satisfies

$$J_0^* J_0 = \Theta_0. \quad (10.5)$$

The strict positivity assumption makes J_0 injective.

It is also surjective. For a nonzero constant vector u , choose x with $Jx = u$. If $\delta = v(x) < 0$, then the leading term of Jx is $t^\delta J_0 x_\delta \neq 0$, a contradiction. If $\delta > 0$, the output cannot have a nonzero constant coefficient. Thus $\delta = 0$ and $J_0 x_0 = u$. The ordinary bounded inverse theorem yields a bounded inverse for J_0 ; (10.5) consequently gives $\Theta_0 \geq \|J_0^{-1}\|^{-2} I$. This proves (ii). $\square$

The argument does not assume that an algebraically bijective adjointable operator automatically has an adjointable inverse. The adjoint of J was constructed directly, and the ordinary inverse theorem was used only after proving that J_0 is onto.

Corollary 10.2 (Coercive equivalent norms need not be isometric [06]). *Let $T = \text{diag}(1/n)$ on ℓ^2 . Then*

$$\Theta = \widehat{T}^2 + q^2 I$$

defines a positive definite form on $\mathcal{H}_\Gamma(\ell^2)$ with

$$q\|x\| \leq \|x\|_\Theta \leq 2\|x\| \quad (x \in \mathcal{H}_\Gamma(\ell^2)), \quad (10.6)$$

yet there is no surjective K -linear isometry from this space to the canonical Hahn–Hilbert space. Both have the same valuation and are spherically complete.

Proof. The lower bound follows from $\langle \Theta x, x \rangle = \|\widehat{T}x\|^2 + q^2\|x\|^2$. The ordinary contraction T remains a contraction after Hahn extension: $I - T^2$ has an ordinary positive square root, so the same identity of positive forms holds over K . Hence $\|x\|_\Theta^2 \leq (1 + q^2)\|x\|^2 < 4\|x\|^2$ for $x \neq 0$. But $\Theta_0 = T^2$ is not uniformly positive, since $\langle T^2 e_n, e_n \rangle = n^{-2}$. Apply Theorem 10.1 and (10.3). $\square$

The operator $T^2 + q^2 I$ is the spectral report’s coercive nonsurjective operator $D^2 + s^2 I$ of `ihshh:thm:coercive`. What the criterion adds is an exact obstruction to *all* bijective linear isometries of the corresponding positive forms, including maps whose regularity was not assumed. At an ordinary uniformly positive residue, positive-order perturbations are removable by (10.4); below that threshold, infinitesimal coercivity does not repair the missing ordinary lower bound.

11 Fredholm perturbations reduce to a finite defect matrix

11.1 The hypotheses and the reduction

Let $T_0 : H \rightarrow G$ be an ordinary bounded Fredholm operator. This means that its ordinary range is closed and that its kernel and cokernel are finite dimensional. Write

$$N = \ker T_0, \quad X = N^\perp, \quad Y = \text{ran } T_0, \quad N_* = Y^\perp = \ker T_0^*.$$

Let $m = \dim N$, $n = \dim N_*$. The restriction $B_0 = T_0|_X : X \rightarrow Y$ is a bounded isomorphism. Zero summands are permitted; identity and inverse formulas on a zero space are interpreted in their unique empty-block sense.

Suppose

$$A = T_0 + E \in \mathcal{B}_\Gamma(H, G), \quad E = 0 \text{ or } v(E) > 0.$$

Relative to $H = X \oplus N$, $G = Y \oplus N_*$, write

$$A = \begin{pmatrix} B & A_{12} \\ A_{21} & A_{22} \end{pmatrix}. \quad (11.1)$$

Here $B = B_0 + \text{positive-order terms}$ is invertible by the Neumann formula, and the other three blocks have positive order or are zero. Define the finite *Schur defect matrix*

$$S = A_{22} - A_{21} B^{-1} A_{12} : \mathcal{H}_\Gamma(N) \longrightarrow \mathcal{H}_\Gamma(N_*). \quad (11.2)$$

Choosing ordinary orthonormal bases identifies it with an $n \times m$ matrix over K . Its entries have positive valuation or are zero.

Theorem 11.1 (Exact finite-defect reduction [03]). *With these hypotheses, the integral block operators*

$$L = \begin{pmatrix} I & 0 \\ -A_{21}B^{-1} & I \end{pmatrix}, \quad R = \begin{pmatrix} I & -B^{-1}A_{12} \\ 0 & I \end{pmatrix} \quad (11.3)$$

have integral inverses, are near the identity, and satisfy

$$LAR = \begin{pmatrix} B & 0 \\ 0 & S \end{pmatrix} =: \mathcal{D}. \quad (11.4)$$

For $r = \text{rank}_K S$,

$$\dim_K \ker A = m - r, \quad (11.5)$$

$$\dim_K(\mathcal{H}_\Gamma(G)/\text{ran } A) = n - r, \quad (11.6)$$

$$\text{ind}_K A = m - n = \text{ind}_\mathbb{C} T_0. \quad (11.7)$$

All identities are exact Hahn identities at arbitrary rank.

Proof. Factor $B = B_0(I + B_0^{-1}(B - B_0))$. Its inverse belongs to $\mathcal{B}_\Gamma(Y, X)$, has integral valuation, and has residue B_0^{-1} . The off-diagonal blocks in (11.3) are positive order. Changing their signs gives the respective inverses of L, R . Direct multiplication first clears the upper-right block by R and then the lower-left block by L , leaving precisely $A_{22} - A_{21}B^{-1}A_{12}$ in the lower-right corner. This proves (11.4).

Because L, R are bijections and B is an isomorphism, the kernel and cokernel of A are isomorphic to those of the finite matrix S . Finite rank-nullity gives (11.5)–(11.7). For support validity, all inverse expansions use the positive monoid generated by the support of E ; multiplication and the finite block decompositions preserve well-ordered support. No analytic or norm-convergent infinite process is involved. $\square$

This is a standard Schur-complement calculation applied in the specified Hahn algebra. Its role is to isolate a finite part where orthogonal linear algebra is safe. In particular, “Fredholm” in (11.7) is the finite algebraic kernel/cokernel condition on these named spaces; it is not a compactoid-operator definition in a different non-Archimedean category.

Remark 11.2 (Comparison with the Hahn Fredholm alternative [merge]). The spectral report’s `ihs:fr:thm:fredholm` treats $I + C$ with C a Hahn series of trace-class operators of nonnegative valuation on one space H . It proves $\text{Det}(I + C) \neq 0 \iff (I + C)^{-1} \in \mathcal{B}_\Gamma(H, H) \iff I + C$ is bijective, that $I + C$ is algebraically Fredholm of index zero, with kernel and cokernel of the same finite dimension, and a factorization of the determinant through one finite matrix. The determinant-free clauses follow from Theorems 11.1 and 12.1: $C = C_0 + C_+$ with C_0 ordinary trace class and $v(C_+) > 0$, so $T_0 = I + C_0$ is an ordinary Fredholm operator of index zero and $A = T_0 + C_+$ satisfies the hypotheses with $m = n$; thus $\dim \ker A = \dim \text{coker } A = m - r$, and if A is bijective then $A^{-1} = A^\dagger \in \mathcal{B}_\Gamma(H, H)$. Theorems 11.1 and 12.1 are more general in these clauses: T_0 may have any index and act between different spaces, and the perturbation need only have bounded, not trace-class, coefficients. They say nothing about the determinant, its finite factorization, or the spectral questions of that report. Source 03 does not make this comparison.

11.2 Finite Moore–Penrose geometry

A finite matrix $S : K^m \rightarrow K^n$ has a Moore–Penrose inverse over K with its standard positive Hermitian form. The finite spectral-theory report (`docs/surcomplex/spectral-theory`) constructs it by the singular value decomposition, proves that the four identities

$$SS^\dagger S = S, \quad S^\dagger SS^\dagger = S^\dagger, \quad (SS^\dagger)^* = SS^\dagger, \quad (S^\dagger S)^* = S^\dagger S \quad (11.8)$$

characterize it, and proves the least-squares theorem `thm:leastssquares`, first over a real closed field; its SVD over a nondivisible Hahn field, `spec:thm:svd`, makes the same arguments available over K for every Γ . Source 03 constructs $S^\dagger$ without the SVD: finite Gram–Schmidt, which needs only the vector-norm roots of [Proposition 3.1](#), gives

$$K^m = \ker S \oplus (\ker S)^\perp, \quad K^n = \operatorname{ran} S \oplus (\operatorname{ran} S)^\perp;$$

the restriction of S between $(\ker S)^\perp$ and $\operatorname{ran} S$ is a bijection of finite-dimensional spaces; invert it there and set it to zero on $(\operatorname{ran} S)^\perp$. Neither construction requires K algebraically closed or F real closed, and no divisibility of Γ is inserted. These finite identities and their geometry are classical Moore–Penrose linear algebra [13], not proposed new results.

12 Exact Moore–Penrose inverses and least squares

12.1 Finite defects restore orthogonal range decomposition

Theorem 12.1 (Fredholm-residue least-squares theorem [03]). *For $A = T_0 + E$ as in [Section 11](#), both $\ker A$ and $\operatorname{ran} A$ are orthogonally split. There is a unique $A^\dagger \in \mathcal{B}_\Gamma(G, H)$ satisfying the four Penrose identities (11.8) with S replaced by A . If P projects onto $\ker A$ and Q onto $\operatorname{ran} A$, it is explicitly*

$$A^\dagger = (I - P) R \begin{pmatrix} B^{-1} & 0 \\ 0 & S^\dagger \end{pmatrix} L Q. \quad (12.1)$$

In particular

$$AA^\dagger = Q, \quad A^\dagger A = I - P. \quad (12.2)$$

Proof. The kernel is finite dimensional by [Theorem 11.1](#), so its orthogonal projection P exists. To prove the exact range assertion rather than assume a Hilbert closed-range theorem, use $A = L^{-1} \mathcal{D} R^{-1}$ from (11.4). Then

$$A^* = R^{-*} \mathcal{D}^* L^{-*}, \quad \ker A^* = L^* \ker \mathcal{D}^*.$$

Here $L^{-*} = (L^*)^{-1}$ and similarly for R . On the finite Schur block, ordinary finite orthogonal algebra over K gives $(\ker S^*)^\perp = \operatorname{ran} S$. On the other block B is invertible. Therefore

$$(\ker \mathcal{D}^*)^\perp = \operatorname{ran} \mathcal{D}.$$

For any subspace Z , the adjoint identity gives $(L^* Z)^\perp = L^{-1} Z^\perp$. Consequently

$$(\ker A^*)^\perp = L^{-1} \operatorname{ran} \mathcal{D} = \operatorname{ran} A. \quad (12.3)$$

The subspace $\ker A^*$ has finite dimension $n - r$, so it is split. Its orthogonal complement (12.3) is split as well; let $Q = I - P_{\ker A^*}$ be its projection. This proves splitting using the finite reduction, not merely finite codimension.

Put

$$A^- = R \begin{pmatrix} B^{-1} & 0 \\ 0 & S^\dagger \end{pmatrix} L.$$

The block identity gives $AA^-A = A$, so AA^- is the identity on $\text{ran } A$. Since $AP = 0$, the map $A' = (I - P)A^-Q$ satisfies $AA' = Q$. Moreover for every x , $A(A^-Ax - x) = 0$, so

$$A'Ax = (I - P)A^-Ax = (I - P)x.$$

Thus $A'A = I - P$. The range of A' lies in $(\ker A)^\perp$, and $A' = A'Q$, which yields $A'AA' = A'$. The two product identities give self-adjointness of AA' , $A'A$ and hence all four Penrose equations. Every factor in A' is adjointable: the finite Schur inverse and finite-rank projections are Hahn operators, and all remaining factors were already constructed in the Hahn algebra.

For uniqueness, any Penrose inverse A'' makes AA'' a self-adjoint idempotent with range $\text{ran } A$, hence $AA'' = Q$. Similarly $A''A$ is a self-adjoint idempotent with kernel $\ker A$, hence $A''A = I - P$. Also $A'' = A''AA''$, so its values lie in $(\ker A)^\perp$. Both $A''f$ and $A'f$ lie there and solve $Ax = Qf$; their difference lies in $\ker A$, and is therefore zero. $\square$

Corollary 12.2 (All least-squares and minimum-norm solutions [03]). *For every $f \in \mathcal{H}_\Gamma(G)$ let $u_0 = A^\dagger f$. Then*

$$\|Ax - f\|^2 = \|(I - Q)f\|^2 + \|A(x - u_0)\|^2 \quad (x \in \mathcal{H}_\Gamma(H)). \quad (12.4)$$

The least-squares minimizers are exactly $u_0 + \ker A$. Among them u_0 is the unique vector of minimum norm. The equation $Ax = f$ is solvable exactly when $(I - Q)f = 0$.

Proof. Write $Ax - f = A(x - u_0) - (I - Q)f$. Its two terms lie in the orthogonal spaces $\text{ran } A$ and $\ker A^*$, respectively. Pythagoras proves (12.4); positivity identifies equality with $A(x - u_0) = 0$. Since $u_0 \in (\ker A)^\perp$, one also has $\|u_0 + k\|^2 = \|u_0\|^2 + \|k\|^2$ for $k \in \ker A$. This proves the minimum-norm and exact-solvability assertions. $\square$

For a finite matrix this is the finite report's `thm:least_squares`; the content here is its transport through the infinite Schur reduction.

Remark 12.3 (Why this does not contradict the graph obstruction [03]). [Example 8.5](#) has an infinite-dimensional ordinary range defect, invisible to a finite matrix: its residue range is dense and not closed. [Theorem 12.1](#) starts with an ordinarily closed Fredholm range and leaves only finitely many kernel and cokernel coordinates to resolve. A finite matrix over K always has the orthogonal decompositions needed here. The two results therefore identify different regimes, not competing definitions of solvability.

13 The exact amplification scale from finite minors

13.1 Finite valuation-ring reduction

The valuation ring $\mathcal{O}$ need not be Noetherian or discretely valued. Nevertheless finite matrices admit the following reduction.

Lemma 13.1 (Finite diagonal reduction over the valuation ring [03]; = `thm:scales`). *Let $S \in M_{n \times m}(\mathcal{O})$ have rank $r > 0$. There are $U_1 \in \mathrm{GL}_n(\mathcal{O})$, $V_1 \in \mathrm{GL}_m(\mathcal{O})$ and exponents $0 \leq s_1 \leq \dots \leq s_r$ such that*

$$U_1 S V_1 = \mathrm{diag}(t^{s_1}, \dots, t^{s_r}, 0). \quad (13.1)$$

If every nonzero entry of S has positive valuation, then $s_1 > 0$. For $1 \leq j \leq r$, set

$$\nu_j(S) = \min\{v(\det S_{I,J}) : |I| = |J| = j, \det S_{I,J} \neq 0\}, \quad \nu_0(S) = 0.$$

Then

$$\nu_j(S) = s_1 + \dots + s_j, \quad (13.2)$$

and the exponents are invariants of S under integral invertible row and column operations.

This is the finite spectral-theory report's theorem “singular scales are determinantal scales”, `thm:scales`, with its invariance lemma `lem:DeltaInvariant`: there $\nu_j = \Delta_j$ and $s_j = \gamma_j$ is the j th singular valuation, and the proof goes through the SVD (`spec:thm:svd` for nondivisible Γ). Source 03 proves it by elimination, without any spectral theorem; the route is kept because it makes the generality of the inverse-scale formula explicit.

Second route by elimination [03]. Choose a nonzero entry of minimum valuation and permute it to the upper-left corner. Write it as $p = t^s u$ with $u \in \mathcal{O}^\times$. Multiplying its row by u^{-1} makes the pivot t^s ; all entries still have valuation at least s . Because the pivot has minimum valuation, it divides every entry in $\mathcal{O}$. Subtract multiples of the first row to clear the first column, then multiples of the first column to clear the first row. These are invertible integral row and column operations. The remaining lower-right block still has all entries of valuation at least s . Repeat on this finite block. The process terminates after r nonzero pivots and produces nondecreasing exponents. This proves (13.1) and the strict positivity assertion. Cauchy–Binet shows that the least nonzero j -minor valuation cannot decrease under an integral left or right multiplication, and applying the integral inverse gives the reverse inequality. For the diagonal matrix in (13.1), the least nonzero j -minor valuation is $s_1 + \dots + s_j$. $\square$

The diagonal exponents in this lemma are valuation invariants, not asserted eigenvalue or singular-value exponents of an arbitrary infinite operator. No spectral theorem is being used for infinite operators.

13.2 A sharp formula for the Moore–Penrose inverse

Theorem 13.2 (Exact finite-defect amplification [03]). *Let $A = T_0 + E$ and S be as in Theorem 12.1. If $r = \mathrm{rank} S > 0$, then*

$$v(A^\dagger) = -s_r = -(\nu_r(S) - \nu_{r-1}(S)) < 0, \quad (13.3)$$

where the s_j are the exponents in Lemma 13.1. If $S = 0$ and $T_0 \neq 0$, then

$$v(A^\dagger) = 0, \quad [t^0]A^\dagger = T_0^\dagger. \quad (13.4)$$

If $S = 0 = T_0$, then $A = A^\dagger = 0$.

Proof. Combine the Schur reduction, multiplication of the regular block by B^{-1} , and the integral finite transformations in [Lemma 13.1](#). There are bounded-coefficient Hahn isomorphisms L on the codomain side and R on the domain side, with integral inverses, such that

$$LAR = \mathcal{E} = \text{diag}(I_X, t^{s_1}, \dots, t^{s_r}, 0). \quad (13.5)$$

To write the regular identity I_X we identify Y with X by B_0 or B as appropriate; this is an isomorphism between the named coefficient summands, not a dimension-changing operation. Every change and its inverse is integral. Zero regular summands are allowed.

Assume first $r > 0$. The diagonal map

$$Z_0 = \text{diag}(I_X, t^{-s_1}, \dots, t^{-s_r}, 0)$$

is an inner generalized inverse of $\mathcal{E}$ and has valuation $-s_r$. Therefore $A_1^- = RZ_0L$ satisfies $AA_1^-A = A$ and has valuation $-s_r$, because integral invertible multipliers preserve operator valuation. The proof of [Theorem 12.1](#) works with any such inner inverse and gives

$$A^\dagger = (I - P)A_1^-Q.$$

Both orthogonal projections are integral by [Lemma 6.3](#). Hence $v(A^\dagger) \geq -s_r$.

For the converse bound put $Z = R^{-1}A^\dagger L^{-1}$. The identity $AA^\dagger A = A$ implies $\mathcal{E}Z\mathcal{E} = \mathcal{E}$. At the j th nonzero finite pivot, its diagonal entry says

$$t^{s_j} Z_{jj} t^{s_j} = t^{s_j}, \quad \text{so} \quad Z_{jj} = t^{-s_j}.$$

A nonzero scalar coordinate of valuation $-s_r$ forces $v(Z) \leq -s_r$. The integral invertible transformations preserve valuation, so $v(Z) = v(A^\dagger)$. This proves equality; the minor formula follows from [\(13.2\)](#).

If $S = 0$, the inner inverse $A_0^- = R \text{diag}(B^{-1}, 0)L$ is integral and its residue is $T_0^\dagger$ relative to the ordinary orthogonal decompositions. Thus $A^\dagger = (I - P)A_0^-Q$ is integral. To identify the projection residues, observe from the Schur formulas with $S = 0$ that

$$\ker A = R(0 \oplus \mathcal{H}_\Gamma(N)), \quad \ker A^* = L^*(0 \oplus \mathcal{H}_\Gamma(N_*)).$$

Because R, L^* and their inverses are near the identity, their integral residue spaces are N, N_* . [Corollary 6.4](#) therefore gives $P_0 = P_N$ and $Q_0 = P_Y$. Taking residues in [\(12.1\)](#) yields $[t^0]A^\dagger = (I - P_N)T_0^\dagger P_Y = T_0^\dagger$. This is nonzero if $T_0 \neq 0$, proving [\(13.4\)](#). If $T_0 = 0$ is Fredholm, both H, G are finite dimensional and the regular block is empty; then $S = A$, so $S = 0$ gives $A = 0$. $\square$

At the finite level, [\(13.3\)](#) for $A = S$ also follows from the finite report: in an SVD $S = U\Sigma V^*$ the unitary factors are integral with integral inverses (`lem:unitaryintegral`), so $v(S^\dagger) = v(\Sigma^\dagger) = -\gamma_r$, which is $-(\Delta_r - \Delta_{r-1})$ by `thm:scales`. The content of [Theorem 13.2](#) is the transport of this formula through the infinite Schur reduction.

Corollary 13.3 (Exactly when the inverse stays finite [03]). *For a nonzero ordinary Fredholm T_0 , the following are equivalent: $A^\dagger$ is integral; $S = 0$; $\dim_K \ker A = \dim_{\mathbb{C}} \ker T_0$. They are also equivalent to preservation of the ordinary cokernel dimension. If even one finite defect direction is lifted, the Moore–Penrose inverse acquires a strictly negative valuation, exactly as measured by [\(13.3\)](#).*

Proof. Use [\(11.5\)](#), [\(11.6\)](#), and [Theorem 13.2](#); all s_j are strictly positive. $\square$

13.3 An off-diagonal term can dominate the inverse scale

Example 13.4 (A fourth-order pole from second- and third-order diagonals [03]). Put $q = t^\eta$, and on $H_0 \oplus \mathbb{C}^2$ take

$$T_0 = I_{H_0} \oplus 0_2, \quad A = I_{H_0} \oplus S, \quad S = \begin{pmatrix} q^2 & q \\ 0 & q^3 \end{pmatrix}.$$

Here H_0 may be infinite dimensional. The finite Schur matrix is exactly S , with

$$\nu_1(S) = \eta, \quad \nu_2(S) = 5\eta, \quad s_1 = \eta, \quad s_2 = 4\eta.$$

Since A is bijective, $A^\dagger = A^{-1}$, and explicitly

$$S^{-1} = \begin{pmatrix} q^{-2} & -q^{-4} \\ 0 & q^{-3} \end{pmatrix}.$$

The inverse valuation is -4η , not -2η or -3η . Off-diagonal coupling is measured by minors and cannot be recovered merely by inspecting the orders on the original diagonal.

More generally, for positive $\alpha, \beta, \gamma \in \Gamma$,

$$S = \begin{pmatrix} t^\alpha & t^\beta \\ 0 & t^\gamma \end{pmatrix} \implies v(S^{-1}) = -(\alpha + \gamma - \min\{\alpha, \beta, \gamma\}).$$

This follows either from the two-minor formula or by writing its inverse. The expression makes sense in any ordered abelian group. For example, in $\mathbb{Q} \oplus \mathbb{Q}$ with lexicographic order take $\alpha = (1, 0)$, $\beta = (0, 1)$, $\gamma = (1, 1)$. Then the inverse valuation is $-(2, 0)$. There is no need to collapse these scales to real numbers or choose an Archimedean embedding.

14 Coherent extension of scales and the full surreal class

14.1 Set-sized scalar extension preserves the formulas

An ordered embedding $\Gamma \hookrightarrow \Delta$ sends a well-ordered support to a well-ordered support, and defines embeddings of scalar fields, vector spaces, and bounded-coefficient Hahn operator algebras. It preserves products, conjugation, adjoints, and the strong sums used in this report. A projection therefore extends to a projection on the larger Hahn space.

Proposition 14.1 (Coherence of the constructive results [03, 06]). *Under any ordered embedding $\Gamma \hookrightarrow \Delta$:*

- (i) *the projection normal form, its direct rotation, the graph formulas and the finite simultaneous straightening extend by coefficient transport;*
- (ii) *for a Fredholm-residue A , its transported Moore–Penrose inverse is the Moore–Penrose inverse of the transported operator;*
- (iii) *the Schur rank is unchanged, its nonzero minor valuations are transported, and the exact amplification formula is unchanged under that transport.*

No cofinality assumption on the embedding is required.

Proof. The normal form and the other constructions in (i) are made from finite algebraic expressions and positive-support binomial series, all preserved by the embedding, since they use exactly the same coefficients and finite contributions at each exponent. The transported Moore–Penrose inverse still satisfies the four Penrose identities on the larger space, since they are operator Hahn identities. Uniqueness in [Theorem 12.1](#) identifies it. A finite determinant is preserved by an injective field map, so the largest nonzero minor size, namely the rank, is unchanged. Its valuation is mapped by the ordered embedding, which preserves minima of finite sets; this proves the final assertion. $\square$

These are algebraic and strong-summation coherence statements. They do not assert that valuation completions or topological closures commute with a noncofinal scale enlargement. The three-duals report explains why those topological operations behave differently.

14.2 A literal vector space over all surcomplex scalars

For readers who want all of $\mathbf{No}[i]$, rather than a specified set subfield, there is a precise class formulation. Work here in von Neumann–Bernays–Gödel class theory without assuming global choice, and keep the ordinary Hilbert spaces H, G as sets. Define

$$\mathcal{H}_{\mathbf{No}}(H) = \left\{ \sum_{\gamma \in S}^H h_{\gamma} t^{\gamma} : S \subset \mathbf{No} \text{ is a set and well ordered, } h_{\gamma} \in H \right\}, \quad t^{\gamma} = \omega^{-\gamma}.$$

Vectors can be encoded as sets of nonzero coefficient pairs. The collection of all these vectors is a proper class when $H \neq 0$. By Conway normal forms the scalar class $\mathbb{C}((t^{\mathbf{No}}))$ is exactly $\mathbf{No}[i]$ in this notation, and $\mathbb{R}((t^{\mathbf{No}}))$ is $\mathbf{No}$ [5]. The vector operations and inner product remain the same coefficient formulas. This gives a compatible system of spaces, not the generally smaller algebraic tensor product of H with the full scalar field.

Every set of vectors and scalars belongs to a common set-sized workspace [06]: collect its exponents and let Γ be the divisible ordered subgroup of $(\mathbf{No}, +)$ generated by them together with 1. The set of finite rational combinations of a set is a set, and this Γ contains all the original supports. The real-closed or algebraically closed Hahn scalar fields can therefore be used locally. All inner-product identities, finite Gram–Schmidt arguments, and positive-order formal series involving finitely or set-many specified inputs can be verified in a common workspace. This supplies a literal positive inner-product vector class over all surcomplex scalars, rather than calling a proper class a set. A separate issue is whether a globally defined operator needs a proper class of coefficient exponents; adjointability gives a strong answer.

Theorem 14.2 (Class extension and set support of adjoints [03, 06]). *In this class formulation the following hold.*

- (i) *The inner product, vector norm, and all support-admissible set-indexed strong sums are well defined over $\mathbf{No}[i]$.*
- (ii) *Every everywhere-defined surcomplex-linear map $A : \mathcal{H}_{\mathbf{No}}(H) \rightarrow \mathcal{H}_{\mathbf{No}}(G)$ with an everywhere-defined class adjoint has a single set-supported Hahn expansion $A = \sum_{\gamma \in S} A_{\gamma} t^{\gamma}$ in ordinary bounded maps $A_{\gamma} : H \rightarrow G$, with $S \subseteq \mathbf{No}$ a well-ordered set. For any set-sized ordered subgroup Γ containing S , A and its adjoint restrict to the corresponding Hahn workspaces.*
- (iii) *Every orthogonal projection, every amplified-graph criterion of [Theorem 8.3](#), and every Fredholm-residue Moore–Penrose construction above applies on these full vector classes. Each resulting operator has its coefficients in one set-sized Hahn workspace, although it acts on vectors from all workspaces.*

Proof. For (i), each operand has set support. The sum, product, and inner product use only finite unions, support sums, and finite coefficient convolutions of those sets. Their validity is the already proved set-sized argument in the subgroup they generate. A set-indexed admissible family has set union of supports by replacement and union; the same reduction applies to its strong sum. The positive-root construction for norms also has set support.

For (ii), restrict A to the set H of constant vectors. Class replacement says that the image $\{Au : u \in H\}$ is a set. Therefore

$$S_A = \bigcup_{u \in H} \text{supp}(Au)$$

is a set of surreal exponents. Do the same for the adjoint restricted to G . Extract coefficients $A_\gamma u = [t^\gamma]A(u)$ as in [Theorem 5.2](#); the ordinary closed graph argument makes them bounded, and [Lemma 5.1](#) makes the nonzero coefficient support well ordered; it is a subset of the set S_A , and the adjoint coefficients are the A_γ^* , with the same support. That proof only uses the two set coefficient spaces and the support sets just exhibited. The series defines an operator $\tilde{A}$ on every global vector, because its support and that vector's support fit in one set-sized subgroup, where Hahn convolution is defined. Finally, for arbitrary x and constant $w \in G$, $\langle Ax, w \rangle = \langle x, A^*w \rangle = \langle \tilde{A}x, w \rangle$, and testing against constants identifies $A = \tilde{A}$ on every class vector. This proves one set-supported expansion for the entire operator, not a separate expansion for each input; convolution with support in Γ preserves the workspace, which proves the restriction assertion.

For (iii), an orthogonal projection is everywhere adjointable, so (ii) applies and the integral-projection and direct-rotation proofs use its set support. For a given surcomplex scalar a choose a set-sized group containing its normal-form support; the residue proof of the graph threshold applies without alteration, as does its explicit inverse in the closed-range case. For a positive-order adjointable perturbation E , part (ii) places all its coefficients in one set-sized group. The Schur and Moore–Penrose formulas can be formed there. Their identities then hold on every class vector: include that vector's support in a larger set-sized group and apply [Proposition 14.1](#). Necessity in the graph obstruction also persists, since any hypothetical global projection would have an ordinarily closed residue by the same argument as before. $\square$

Source 06 states the descent (its Theorem 11.1) in NBG *with* global choice. Its proof, like source 03's, uses choice only on the sets H and G (the dependent choice in [Lemma 5.1](#) and the ordinary closed graph theorem), together with class replacement; global choice is not needed, and the theorem is stated here without it. The quantifier is over a *given* class map and its given class adjoint; it does not require a set containing all class maps, and Zorn's lemma is not applied to a proper class of subspaces. The ordinary set of constants is crucial: it converts initially class-valued operator data into one set of possible exponents before the support argument begins.

Corollary 14.3 (Global geometric classification and persistent defects [06]). *Every global orthogonal projection and every global unitary descends to a set-sized Hahn workspace. The residue, graph and unitary classification of [Sections 6](#) and [7](#) holds over literal **No** and **No**[i]. The unbounded-graph counterexamples, the finite nonlattice obstruction, the missing countable join, and the metric rigidity criterion also hold globally.*

Proof. An orthogonal projection is its own everywhere-defined adjoint; a unitary has its inverse as adjoint. Apply [Theorem 14.2\(ii\)](#) and perform the finite algebraic and positive-order formal constructions in a workspace containing the resulting support. For unbounded graphs the adjoint-domain coefficient proof is unchanged globally. If such a graph had a global orthogonal projection, its ordinary residue would be closed, contradicting the same domain computation. The finite-meet

and countable-join proofs use only that residue fact and the existence of rank-one projections, so they also apply without alteration. In the metric proof, a hypothetical bijective isometry again has adjoint ΘJ^{-1} ; use [Theorem 14.2\(ii\)](#) in place of [Theorem 5.2](#) and repeat the residue argument. $\square$

If a global split subspace is $M = \text{ran } P$ and Γ contains $\text{supp } P$, then

$$M \cap \mathcal{H}_\Gamma(H) = \text{ran}(P|_{\mathcal{H}_\Gamma(H)}). \quad (14.1)$$

The nontrivial inclusion follows from $m = Pm$ for $m \in M$. Consequently the global range is the compatible union of its workspace ranges [06].

This extension explains why the operator theorems genuinely concern surcomplex scalars, not only a convenient notation for ordinary series. It does not turn the global vector class into an ordinary set-sized Hilbert space. Nor is a “class of all subspace classes” formed. Orthogonal projections can instead be represented by their set Hahn codes; the non-meet example of [Theorem 9.2](#) still obstructs the corresponding projection-code order in infinite coefficient dimension.

14.3 Global fine topology: set-sized discreteness

On a global space use the fine norm neighborhoods $\{x : \|x - a\| < \varepsilon\}$ for all positive surreal ε .

Proposition 14.4 (Set-sized discreteness in the fine topology [03, 06]). *In $\mathcal{H}_{\mathbf{No}}(H)$, every set of vectors is closed and discrete in the fine norm topology, and its distinct points have a common positive surreal separation bound. A net indexed by a set that converges in the fine topology is eventually equal to its limit, and one that is Cauchy for all positive surreal radii is eventually constant. Each fixed Hahn workspace, viewed as a subset of the global space, is therefore discrete in the induced fine topology.*

Proof. This is the vector form of the analysis report’s `a:thm:discrete`, and of the foundations report’s `found:thm:discrete`, whose surcomplex clauses are formalized in Lean according to `docs/FORMALIZATION.md`; neither source cites them. The mechanism is the surreal cut property. For a set X of vectors, the nonzero distances between distinct members form a set $D \subseteq \mathbf{No}_{>0}$, and the cut $\{0 \mid D\}$ gives $0 < \varepsilon < d$ for every $d \in D$ (any positive number if D is empty). Thus each point of X is isolated. For $a \notin X$, apply the same argument to $\{\|a - x\| : x \in X\}$; a ball around a misses X , so X is closed. For a set-indexed net converging to a , apply the separation argument to its range together with a : eventual membership in the resulting ball around a is eventual equality to a . For a Cauchy net, the nonzero pairwise differences form a set of vectors; choose a positive radius below all their norms, and the Cauchy condition at that radius makes a tail constant [03]. A workspace is a set, so the last assertion follows. $\square$

When $\Gamma \neq 0$ and $H \neq 0$, the *intrinsic* valuation topology on $\mathcal{H}_\Gamma(H)$ is not discrete: every valuation neighborhood of zero contains nonzero monomial vectors, and nonconstant sequences such as $t^n \rightarrow 0$ converge inside $\mathbb{C}((t^{\mathbb{Q}}))$. Those sequences only pass the *workspace* tests; in the full surreal field, a radius smaller than all their nonzero terms prevents convergence. Thus the spherically complete set-valued model, the class-valued fine model, and strong formal summation should not be conflated. The global operator theorem is algebraic; it neither asserts useful set-indexed fine convergence nor needs it. No claim is made about arbitrary class-indexed Cauchy families.

15 The report in the collection, the literature, and open questions

15.1 Relation to neighbouring reports

The following repository results are predecessors, not contributions of this report.

- *Infinite-Dimensional Hahn Spectral Theory* (`docs/surcomplex/infinite-dimensional-hahn-spectral-theory`). Part I constructs $H((t^\Gamma))$ and proves the positive inner product (`ihs:hh:prop:inner`), the automatic adjoint theorem and bound (`ihs:hh:thm:adjoint`, `ihs:hh:cor:autobounded`), the norm-attainment criterion (`ihs:hh:thm:norm`), positive-order defect rigidity (`ihs:hh:thm:defect`), the coercive operator $D^2 + s^2I$ (`ihs:hh:thm:coercive`) and a closed proper range with zero orthogonal complement (`ihs:hh:thm:closedrange`). Part IV proves the support-controlled direct rotation (`ihs:fr:lem:rotation`) and the Hahn Fredholm alternative (`ihs:fr:thm:fredholm`). This report uses the first group as background, derives its normal form from the rotation lemma (Theorem 6.5), and contains the determinant-free clauses of the Fredholm alternative (Remark 11.2). It does not reproduce the spectral classifications, claim a new discovery of nonrepresented continuous functionals, or rename a Fredholm determinant as a Moore–Penrose inverse. The spectral report’s Part I remarks that one “could change the vector construction ... or impose a stronger orthogonal-completeness axiom”; by Solèr’s theorem (Section 15.2) the stronger axiom cannot be orthomodularity itself for a set-sized space over K with an infinite orthonormal sequence, and Theorem 8.7 exhibits an explicit orthoclosed witness. No construction is changed here. The missing countable join (Theorem 9.5) does not contradict that report’s complete Boolean algebra of coordinate projections (`ihs:rf:thm:commutant`), which concerns constant coordinate projections in its row- and column-finite model; the P_n here are not constant.
- *Three Duals at Surreal Scales* (`docs/surcomplex/three-duals-of-hahn-vector-spaces`): spherical completeness (`duals:prop:spherical`), the coefficientwise Riesz criterion (`duals:thm:riesz`), the closed hyperplane with zero orthogonal complement (`duals:thm:hyperplane`) and the exact distance cut (`duals:thm:distance`), which Proposition 8.9 repeats for an orthoclosed subspace. That report also distinguishes algebraic scalar extension, valuation completion and strong closure.
- *Spectral theory* (`docs/surcomplex/spectral-theory`), the finite report: finite Moore–Penrose inverses and least squares (`thm:least squares`), singular scales as determinantal scales (`thm:scales`, `lem:DeltaInvariant`) and the SVD over a nondivisible Hahn field (`spec:thm:svd`); Lemma 13.1 and the finite part of Section 11 are these results [19]. Its non-claim that it has “no theorem ... about infinite-dimensional Hilbert spaces over $\mathbf{No}[i]$ ” is continued by the spectral report and by this one.
- *Analysis and Foundations*: fine discreteness of sets (`a:thm:discrete`, `found:thm:discrete`), of which Proposition 14.4 is the vector form [20, 21].
- *Hidden negative directions* (`docs/surcomplex/hidden-negative-hermitian-directions`): its two-scale positivity criterion for $P + \varepsilon Q$ is related in spirit to Theorem 10.1; neither report answers a question of the other.

The relevant repository portions were inspected by the sources, not the complete proof corpus: source 03 read the relevant parts of the spectral and duals reports; source 06 read the root README, the opening of the formalization ledger, the spectral report’s README and article

opening, and the duals report’s README. Neither source claims inherited Lean verification, and docs/FORMALIZATION.md maps no label of this report to Lean.

15.2 Primary literature and the novelty boundary

Both literature searches were targeted, not exhaustive. Source 03 searched the phrases “Hahn Hilbert orthogonal graph”, “Hahn Moore–Penrose”, “Hahn Fredholm least squares”, non-Archimedean orthogonal complementation and the works below; source 06 searched Hahn/Hilbert orthogonal graphs, projection lattices and isometric positive forms. Failure to identify an exact predecessor is evidence for a candidate contribution, not a proof that none exists under other terminology.

Solèr’s theorem [06]. For a Hermitian space E call $M \subseteq E$ orthoclosed if $M = M^{\perp\perp}$, and call E orthomodular if $E = M \oplus M^{\perp}$ for every orthoclosed M . Solèr’s theorem says that an orthomodular Hermitian space with an infinite orthonormal sequence is a classical real, complex or quaternionic Hilbert space; see Holland’s exposition, Theorem 1.3 [7, 16]. Consequently a nonclassical scalar field such as F or K cannot retain this entire package. The theorem applies to set-sized scalar fields and spaces; it is not applied here to proper classes. An abstract prediction that some Hilbert axiom must fail is therefore not an original contribution; the proposed contribution is the explicit infinitesimal graph mechanism, its exact boundedness criterion, its polynomial positive-kernel realization, and the concrete failure of binary lattice operations and of a countable join. Modern categorical consequences of Solèr’s theorem are actively studied; for example Lack and Tobin [10] characterize the category of ordinary Hilbert spaces.

Non-Archimedean Hilbert-like spaces. Aguayo–Nova study non-Archimedean Hilbert-like c_0 spaces and orthogonal-complementation obstructions [1]. Their model is not the layered coefficient space retaining all of an ordinary Hilbert H . Source 03’s comparison used the publication record and available author-posted text; a blocked publisher download was not treated as a completed full-text audit. Ishizuka [8] studies compactoid self-adjoint spectral theory over a non-Archimedean field with formally real residue field; those hypotheses and that Banach-space category differ from the coefficient Hahn space with an order-valued inner product, and a result in either setting is not automatically a theorem in the other. Neither source claims a first non-Archimedean Hilbert space, a first failure of Riesz representation, or a first non-Archimedean spectral theorem.

Projections and perturbations. Bursztyn–Waldmann develop formal deformations of Hermitian vector bundles and finite projective modules, including uniqueness up to isometric equivalence [4]. Simon discusses unitaries relating projections and records the classical direct-rotation formula [14, 15]; Kato’s book is the standard source for the two-projection theory and for closed operators [9]. Neither projection deformation nor that formula is claimed to originate here. Avrachenkov–Lasserre investigate analytic perturbations of generalized inverses [2], and Müller–Strohmaier develop Hahn-meromorphic functions and a holomorphic Fredholm theorem [11]. They are important analytic precedents. The present proofs use instead unrestricted well-ordered supports over an arbitrary ordered group, finite-defect orthogonal geometry, and exact leading valuation from minors; their analyticity and convergence hypotheses are not imported, and neither their methods nor classical generalized inverses are presented as new. Source 03’s comparison with [2] used its abstract and the coauthor’s publication description, not the full text.

Foundations. Conway supplies the normal-form and small-cut foundations [5]; Neumann supplies the classical support mechanism [12]. Formal infinite-sum spaces and strong linearity have an established contemporary literature, including [3, 6]; those constructions and that terminology are not new here.

Table 1: Attribution and the boundary of the proposed contributions, as stated by the sources.

Result here	Prior input	Candidate contribution or status
Construction, positivity, completeness, automatic adjoints, norm bounds	Spectral and duals reports	Reproved background; not new
Residue normal form, graph chart	Automatic adjoints; direct rotation (<code>ihs:fr:lem:rotation</code>)	Global classification of split subspaces in this model (03, 06)
Unitary group, orbits, finite straightening	Formal exponential calculus; two-projection intertwiner	Exact statements in this Hilbert setting (06)
Amplified graphs	Normal equations; ordinary closed-range inverse	Exact finite/infinite threshold and residue obstruction (03)
Infinitesimal unbounded graphs; positive kernel; countable join	Closed-operator adjoints; Solèr's theorem	Explicit orthoclosed mechanism and its consequences (06)
Scaled closed relations	The two graph theorems	Added by the merge
Split-subspace poset	Orthomodular identities; finite projection theorem	Lattice if and only if coefficient dimension is finite (03; one direction 06)
Metric rigidity	Coercive operator of <code>ihs:hh:thm:coercive</code>	Exact ordinary-residue threshold (06)
Fredholm least squares	Schur reduction; finite Penrose geometry (<code>thm:leastquares</code>)	Arbitrary-rank existence and exact infinite-space solutions (03)
Sharp inverse valuation	Finite valuation-ring elimination; <code>thm:scales</code>	Exact finite-Schur minor formula for the full pseudoinverse (03)
Full surcomplex extension	Class replacement; normal forms; automatic adjoints	Explicit set support of global adjointable operators (03, 06)

Several positive components are natural extensions of classical formulas, so their priority should be assessed at the level of the precise hypotheses and combined classification, not the formula alone. Independent mathematical review and a wider bibliographic check remain necessary before any publication claim of priority.

15.3 Open questions

The sources' questions are collected here, with what the other source settles. None is a named published problem.

Question 15.1 (Beyond Fredholm residues [03]). The Fredholm assumption is sufficient, not asserted necessary, for an adjointable Moore–Penrose inverse. Which operators in $\mathcal{B}_r(H, G)$ have one? A broader characterization would have to treat infinitely many interacting defect directions without assuming that their inverses share a well-ordered support of bounded coefficient operators; the graph threshold ([Theorem 8.3](#)) is a precise obstruction any such characterization must explain.

Question 15.2 (Orthoclosed subspaces with closed residue [03, 06]). Source 03 asks when an independently presented valuation-closed subspace admits a positive-order graph chart, and makes no claim that closed residue suffices. [Proposition 6.7](#) (completing a remark of source 06) settles that negatively for valuation-closed subspaces, with a subspace that is not orthoclosed. For orthoclosed subspaces the question is open; within the scaled relations of [Theorem 8.10](#), closed residue is equivalent to splitting. Source 06 asks, more specifically, which kernels of regular Hahn operators

have closed residues and which higher-order conditions make them complemented; the graph chart gives a local model, but closed residue is not asserted to settle every orthoclosed subspace.

Question 15.3 (Infinite orthogonal synthesis [06]). For infinite orthogonal families, a coherent common support and bounded ordinary coefficient synthesis are separate necessary conditions (Corollary 9.6 and Proposition 9.7). Which natural sufficient conditions, stable under composition and scale enlargement, make a countable family simultaneously straightenable, or give it a join?

This bears on the first further question of the spectral report’s Part IV (`ihs:fr:sec:questions`): what uniform spectral-gap and coefficient-ideal hypotheses ensure that all cluster rotations assemble into a bounded-coefficient Hahn unitary? Assembling the cluster rotations into one unitary would conjugate all the cluster projections simultaneously, so the two conditions above are necessary for it; Corollary 9.6 and Proposition 9.7 give families for which each fails. No sufficient hypothesis is proved, and that question stays open, as do its other two parts (regularized spectral products; compact residues with a kernel), which neither source addresses.

Question 15.4 (Singular positive forms [06]). Theorem 10.1 suggests a stratification of positive Hahn metrics by the ordinary lower-bound behaviour of their leading coefficient, with more singular forms treated by a different coefficient space rather than by a forced canonical isometry.

Question 15.5 (Structures not developed [03]). Source 03 lists general tensor products, frame bounds, spectral measures, unbounded operators with proper domains, and infinite-family projection joins as not developed. Within this report source 06 treats two of them: graphs of closed unbounded operators (Theorem 8.7), and countable projection joins, negatively (Theorem 9.5: a countable orthogonal family need not have a join). Which countable families do have joins is not decided. Tensor products, frame bounds and spectral measures remain undeveloped; by Theorem 9.5 no construction can assign the join of *every* countable orthogonal family, but this does not forbid spectral measures for particular operators or other coefficientwise summation rules.

For computational use, the finite Schur matrix is exact but its entries may themselves be infinite Hahn series. The minor formula specifies an invariant and an algebraic reduction; it is not an algorithm for deciding equality or leading terms of arbitrary uncomputably presented surreal numbers. Effective implementations need an explicit support and coefficient representation.

15.4 Proof status

All central arguments are given in full above or in the cited reports. Their main dependencies are the ordinary Hilbert closed-graph, Baire, bounded-inverse and Riesz theorems, the adjoint theorem for closed densely defined operators, ordinary positive square roots, the Hahn–Neumann support lemma, Conway normal forms, and finite linear algebra; these are identified inputs, not proved from set theory here. The class section additionally uses class replacement in NBG without global choice. No proof assistant was run, neither source has been independently refereed, and no assertion rests on numerical evidence. The mathematical claims should be evaluated by checking the proofs and their hypotheses, not the novelty wording.

A Provenance

A.1 Section by section

Table 2: Provenance of each part of the report. Source section and theorem numbers refer to the manuscripts, which are not shipped.

Part of this report	Sources and choices
Section 1	The merge, from 03 §1 and 06 §1 and abstracts.
Section 2	03 §2; 06 §§2.1–2.2 (support lemma, strong summability, the lexicographic example). Neumann inverse and binomial series cited from the spectral report.
Section 3	03 §3 and 06 §§2.3, 3 printed once: the positive form (03 Prop. 3.1, 06 Prop. 2.2) and spherical completeness (03 Thm. 3.2, 06 Prop. 3.1) by citation of <code>lhs:hh:prop:inner</code> and <code>duals:prop:spherical</code> with sketches; 06 Prop. 3.3 (the c_{00} example); 03 Prop. 3.4.
Section 4	03 §4 and 06 §2.4 printed once; 06 §5: Prop. 5.1 kept, Prop. 5.2 by citation of <code>duals:thm:riesz</code> , Thm. 5.3 by citation of <code>lhs:hh:thm:norm</code> with a rectangular reduction [merge].
Section 5	03 §5 and 06 §4: automatic adjoints (03 Thm. 5.1, 06 Thm. 4.2, Cor. 4.3) by citation of <code>lhs:hh:thm:adjoint</code> and <code>lhs:hh:cor:autobounded</code> , with the reduction of the rectangular case [merge]; 06 Lem. 4.1 (Baire) and §4.1 (calculus) kept; 03 §5.2.
Section 6	03 §6 and 06 §6 printed once: 03 Prop. 6.2; 03 Lem. 6.3 = 06 Lem. 6.1 with both proofs; 03 Cor. 6.4; 03 Thm. 6.5 = 06 Thm. 6.2 derived from <code>lhs:fr:lem:rotation</code> ; 03 Cor. 6.6 = 06 Thm. 6.3. Proposition 6.7 completes 06’s remark after its Thm. 6.3 and answers 03 Rem. 6.7 [merge].
Section 7	06 Cor. 6.4 and §7 (Thms. 7.1, 7.2).
Section 8	03 §7 (Prop. 7.1, Thm. 7.2, Rem. 7.3, Ex. 7.4, Cor. 7.5); Lemma 8.2 is 06 Lem. 9.2 and the step in 03’s proof of its Thm. 8.2; 06 §8 (Thm. 8.1, Cor. 8.2, Prop. 8.3, the last credited to <code>duals:thm:distance</code>); Section 8.4 is the merge’s, from an observation of the placement dossier.
Section 9	03 §8 (Prop. 8.1, Thm. 8.2, Rem. 8.3), which contains 06 Thm. 9.3; 06 §9 (Thm. 9.1 in 03’s coordinates, the Rickart remark, Thm. 9.4, Cor. 9.5, Prop. 9.6).
Section 10	06 §10.
Sections 11 to 13	03 §§9–11. The comparison with <code>lhs:fr:thm:fredholm</code> is the merge’s; the finite Moore–Penrose paragraph and Lem. 11.1 are cited from the finite spectral-theory report, 03’s elimination proof kept as a second route.
Section 14	03 §12 and 06 §11 printed once: 03 Prop. 12.1 with 06’s coherence remarks; 03 Thm. 12.2 = 06 Thm. 11.1, stated without global choice; 06 Cor. 11.2 and (14.1) ; 03 Prop. 12.4 and 06 Prop. 11.3 printed once as the vector form of <code>a:thm:discrete</code> .
Section 15	03 §13 and 06 §12, Appendix A; the relations to the collection and the status of the questions are the merge’s.
Section C	03 Appendix A, with rows for 06’s results and the merge.

Where the sources overlap, the statement with the weakest hypotheses is printed. The proofs attached to printed statements were checked for hidden uses of stronger hypotheses: 06’s descent proof uses choice only on sets, so the statement without global choice (03’s) is correct for both; 03’s

no-meet proof and its transfer to an arbitrary infinite-dimensional H use no property of ℓ^2 beyond containing three orthogonal copies of it.

A.2 Labels

Every label carries the prefix `hgeo:.` The placed text was source 03 with 69 bare labels; nothing in the collection cited them, and each was prefixed unchanged (for example `thm:threshold` is `hgeo:thm:threshold`); the other labels were added for source 06’s results, the merge, and the appendices.

A.3 Renamed symbols

Table 3: Symbols renamed from the sources.

Source	Source symbol	Here, and why
06	$K = \mathbb{R}((t^\Gamma)), F = k((t^\Gamma))$	F and K swapped back: F real, K complex, as in 03, the spectral report and the finite spectral-theory report; k is $\overline{\mathbb{Q}}$.
06	$\mathcal{E}_\Gamma(H), \mathcal{E}_{\mathbf{No}}(H)$	$\mathcal{H}_\Gamma(H), \mathcal{H}_{\mathbf{No}}(H)$ (03’s), because $E_\Gamma(V)$ is the algebraic extension $\text{span}_K V$ in the three-duals report.
06	$\mathcal{A}_\Gamma(H, G), \mathcal{A}_{>0}, v_{\text{op}}$	$\mathcal{B}_\Gamma(H, G), \mathcal{B}_\Gamma^{>0}(H, G), v$ (03’s).
06	$\mathcal{U}(H), \mathcal{U}_{1,\Gamma}(H)$	$\mathbf{U}(H), \mathbf{U}_{1,\Gamma}(H)$, since the spectral report’s $\mathcal{U}(T)$ is a set of norm bounds.
06	$\tau = t^\delta$	$q = t^\eta$ (03’s); $\delta \mapsto \eta$ throughout.
06	S ($Se_n = e_n/n$), D ($\text{diag}(n)$)	T (03’s) and $\Lambda = T^{-1}$; the spectral report writes D for $\text{diag}(1/n)$, and D was also 03’s rotation denominator.
06	$M_\tau(A)$	$M_q(A); M_b(R)$ for relations (merge).
06	$T(x, y) = \tau x - Sy, L = T^*T$	$B(x, y) = Tx - qy$ (03’s, input coordinates exchanged) and $L = B^*B$.
06	R_n	P'_n , since R is a relation.
06	$D = P - Q,$ $A = PQ + (I - P)(I - Q),$ $Q = \widehat{P}_0$	written out, W (03’s), P_0 .
06	$A = \sum P_j Q_j$ (finite straightening)	W .
06	$G = \widehat{G}_0 + R$ (metric)	$\Theta = \widehat{\Theta}_0 + \Theta_+$, since G is a Hilbert space and R a relation; R' is Θ' .
06	h in its Thm. 6.2, Cor. 8.2	u, h' ; $h = (1/n)$ is reserved.
06	D (c_{00})	c_{00} .
06	“projector”, “orthogonally complemented”	“projection”, “split” (both words are used; they are synonyms).
03	$D = I - (P - P_0)^2$	written out.
03	$C = \ker T_0^*$ and blocks C_1, D, F_1	N_* and A_{12}, A_{21}, A_{22} , since F is the real field and D was overloaded.
03	$G = Y \oplus C$ in the threshold proof	$G = Y \oplus Y^\perp$.
03	G_0, F, F' (Penrose proof)	A^-, A', A'' , since G is a Hilbert space and F the field.

Source	Source symbol	Here, and why
03	J, K_1, G (scale proof)	$L, R, A_1^-; A_0^-$ in the case $S = 0$.
03	U, V (Lemma 11.1)	U_1, V_1 , since U is a unitary.
03	$\eta = v(b)$ in the threshold proof	η' , since η is fixed by $q = t^\eta$.
03	$R = (I + X^*X)^{-1}$ in the graph chart	R_X .

A.4 Stale and incomplete repository statements

The pins 9a385d3 (03) and a45d722 (06) are kept as provenance. None of the cited reports changed between the pins and the placement, so the corrections below concern omissions of the sources rather than later changes of the collection.

1. Neither source cites the rotation lemma `ih:fr:lem:rotation`, which already states the direct rotation for every projection P of the algebra with $v(P - P^\circ) > 0$; 03 says only that Part IV “uses the classical direct rotation” for clusters. Given the lemma, the shared normal-form theorem is a corollary of the integral-projection lemma and the automatic adjoint theorem ([Theorem 6.5](#)).
2. Source 03 does not compare its Fredholm theorems with `ih:fr:thm:fredholm`; [Remark 11.2](#) does.
3. Source 03 does not cite the finite spectral-theory report, whose `thm:least_squares`, `thm:scales` and `lem:DeltaInvariant` are its finite Moore–Penrose paragraph and its Lemma 11.1. 03’s remark that its finite construction needs neither real closedness nor divisibility is correct, but at 03’s pin that report already proved the SVD over a nondivisible Hahn field (`spec:thm:svd`), so the finite results were available for every Γ .
4. Source 06’s distance cut (its Prop. 8.3) does not credit `duals:thm:distance`, which has exactly the same cut for the closed hyperplane ([Proposition 8.9](#)).
5. Neither source cites `a:thm:discrete` (present at both pins) or the Lean-formalized `found:thm:discrete` for fine discreteness ([Proposition 14.4](#)).
6. Source 06 states its global descent theorem in NBG with global choice; the proof does not use it ([Theorem 14.2](#) and the paragraph after it).
7. Source 06 cites the spectral report by theorem numbers (6.1, 11.3, 13.2) and the duals report by Proposition 5.1, Proposition 8.1 and Sections 8–9; these numbers are correct at the current build and are replaced here by the labels `ih:hh:thm:adjoint`, `ih:hh:thm:coercive`, `ih:hh:thm:norm`, `duals:prop:spherical`, `duals:prop:norm`, `duals:thm:riesz`. Its description of the spectral report as 99 pages is still that report’s page count.
8. Source 03 lists unbounded operators with proper domains and infinite-family projection joins as not developed, and makes no claim that closed residue suffices; within this report source 06 treats the first two (the second negatively), and [Proposition 6.7](#) settles the third negatively ([Questions 15.2](#) and [15.5](#)).
9. Source 06’s non-lattice theorem is stated for ℓ^2 only; 03’s theorem covers every infinite-dimensional H and adds the finite-dimensional converse ([Theorem 9.2](#)).
10. Both sources describe their delivery archives (source PDFs, build records). The source manuscripts and PDFs are not shipped with this report; the shipped files are listed in [Section D](#).

B What this report does not claim

Every limitation, disclaimer and priority caveat stated by a source is kept here, numbered per source.

B.1 Source 03 (34 items)

- 03.1 An AI-assisted research draft, not independently refereed or formally verified; no proof assistant was run, and there is no inherited Lean verification.
- 03.2 The construction, automatic adjoints, the direct-rotation formula and the classical perturbation tools are credited, not claimed as new.
- 03.3 No invention is claimed of non-Archimedean inner products, non-Archimedean failures of Riesz representation or orthogonal complementation, formal deformation of projections, the Schur complement, or the Moore–Penrose inverse.
- 03.4 The proposed contributions (the exact graph threshold, the finite-dimensional/lattice dichotomy, the combined finite-defect least-squares and sharp-valuation package) were not located by a targeted comparison; priority is not certified, and failure of the searches is evidence for a candidate contribution, not proof that none exists.
- 03.5 No named published conjecture is asserted to be solved.
- 03.6 Finite symbolic checks are supplementary, not proofs; they can catch sign or transpose mistakes but do not establish support well-ordering, infinite-dimensional boundedness, spherical completeness, the class argument, or bibliographic priority.
- 03.7 The model does not retain ordinary Hilbert-norm convergence as the topology on the constant copy of H .
- 03.8 The classification is of orthogonally split subspaces, not of all valuation-closed subspaces; near-identity unitaries representing a split subspace are not unique.
- 03.9 Positive-order power series are identities of Hahn series and do not presuppose topological convergence of finite partial sums.
- 03.10 Common workspaces for specified sets of scalars do not reduce every class-sized problem to a single set field, nor identify the full fine topology with the workspace topology.
- 03.11 Not every positive element of F has a square root; only the roots of squared norms are asserted.
- 03.12 The rank-one ultrametric $\exp(-v(x - y))$ is not the F -valued Hilbert norm.
- 03.13 Spherical completeness does not imply the projection theorem and does not force ordinary completeness on coefficient subspaces.
- 03.14 The constant orthonormal basis need not be a valuation Schauder basis; Parseval holds coefficientwise only.
- 03.15 Neither Riesz observation is claimed as new.
- 03.16 The automatic adjoint theorem does not classify all valuation-continuous maps, and uses ordinary Hilbert completeness of the coefficient spaces.
- 03.17 No general bounded-inverse theorem over K is assumed.

- 03.18 The rotation formula is not a new result.
- 03.19 Closed residue is necessary, not claimed sufficient, for splitting.
- 03.20 Positivity of $I + A^*A$ does not alone imply surjectivity.
- 03.21 The no-nearest-point example asserts failure of an attained minimum only, not a statement about a distance infimum.
- 03.22 The orthomodular-poset axioms do not assert that arbitrary binary meets or joins exist; the general non-Archimedean nonorthomodularity phenomenon is not new [1].
- 03.23 Nontrivial Γ is essential to the lattice dichotomy.
- 03.24 “Fredholm” is the finite algebraic kernel/cokernel condition, not a compactoid-operator notion; the Schur calculation is standard; the finite Moore–Penrose identities are classical.
- 03.25 The diagonal exponents are valuation invariants, not asserted eigenvalue or singular-value exponents of an arbitrary infinite operator; no spectral theorem is used to diagonalize an arbitrary Hahn operator.
- 03.26 The coherence statements are algebraic and do not assert that valuation completions or closures commute with a noncofinal scale enlargement.
- 03.27 The class section does not make the global vector class an ordinary set-sized Hilbert space, and forms no class of all subspace classes.
- 03.28 The fine-net result concerns set-indexed nets only; no claim is made about class-indexed Cauchy families.
- 03.29 The Fredholm assumption is sufficient, not asserted necessary, for an adjointable Moore–Penrose inverse.
- 03.30 General tensor products, frame bounds, spectral measures, unbounded operators with proper domains and infinite-family projection joins are not developed in source 03 (see [Question 15.5](#) for what source 06 adds); no classical infinite-dimensional spectral theorem, measurement postulate or physical interpretation is claimed from the existence of a positive surreal norm.
- 03.31 The minor formula is not an algorithm for uncomputably presented surreal numbers.
- 03.32 The repository comparison inspected the relevant portions only, not the complete proof corpus or all unintegrated manuscripts; no novelty is claimed relative to uninspected sources.
- 03.33 The comparisons with Aguayo–Nova and Avrachenkov–Lasserre were not full-text audits (publication record and author text; abstract and coauthor’s description); neither projection deformation nor the rotation formula originates here, and neither the analytic methods of [2, 11] nor classical generalized inverses are presented as new.
- 03.34 No theorem of the main set-sized treatment assumes a divisible group; no ordinary coordinate sum is exchanged with a strong Hahn family without qualification; nearest-point statements compare exact norms and do not replace minimization by an unproved infimum.

B.2 Source 06 (33 items)

- 06.1 An AI-assisted research draft with complete written proofs; not independently refereed or proof-assistant verified; no surreals-specific theorem is labelled Lean-verified.

- 06.2 Candidate contributions are separated from repository results and classical operator theory; the literature comparison is targeted, not a certification of priority.
- 06.3 No named published conjecture or open problem is claimed solved.
- 06.4 “Candidate” qualifies originality only, not the logical status of the proofs.
- 06.5 Inherited, not claimed: the construction, automatic adjointability, the failure of Riesz representation, the norm-attainment criterion, spherical completeness, and the distinctions among represented, strong and continuous duals.
- 06.6 The Kato–Nagy intertwiner formula is emphatically not new; the formula is canonical as a formula, not the only near-identity unitary.
- 06.7 Solèr’s theorem is applied only to set-sized fields and spaces, not to proper classes; an abstract prediction from it is not an original contribution.
- 06.8 The normal-form framework is classical; divisibility is used only when mentioned.
- 06.9 The positive-word support theorem is imported, not inferred from convergence.
- 06.10 Strong summability is neither the strong operator topology nor ordinary convergence; formal summability has an established literature.
- 06.11 The space is larger than the algebraic tensor product; no density of the algebraic scalar extension is assumed.
- 06.12 The valuation topology is that of a fixed workspace, not the topology of all positive tolerances in the full surreal field.
- 06.13 Spherical completeness concerns valuation balls only, not arbitrary ordered-field norm balls, and implies neither ordinary completeness of the coefficient space nor complementation of closed subspaces.
- 06.14 The c_{00} example is not orthoclosed and is not a witness to the failure of orthomodularity.
- 06.15 Positive-order operator series are not justified by $n\delta \rightarrow +\infty$; formal smallness is not a real operator-norm estimate.
- 06.16 The order bound is not a construction of an F -valued Banach C^* -algebra norm.
- 06.17 “Chart” asserts no differential or manifold structure.
- 06.18 Closed residue alone is insufficient; the bounded-coefficient hypothesis is part of the classification.
- 06.19 No real norm convergence of the Baker–Campbell–Hausdorff law is asserted; formal exponential correspondences are established material.
- 06.20 The finite qualification in simultaneous straightening is essential.
- 06.21 The metric proof does not assume that an algebraically bijective adjointable operator has an adjointable inverse.
- 06.22 The global theorem quantifies over a given class map and its given class adjoint; no set of all class maps is formed and Zorn’s lemma is not applied to a proper class of subspaces.
- 06.23 The global operator theorem is algebraic; it neither asserts useful set-indexed fine convergence nor needs it.

- 06.24 Workspace coherence is algebraic and strong-summation coherence, not a claim that intrinsic completions commute with arbitrary enlargement.
- 06.25 “Hahn–Hilbert” names a construction with selected Hilbert features; it does not assert orthomodularity or the full classical package.
- 06.26 The spectral report’s spectra are not reproved or expanded, and are not identified with a real Banach-algebra spectrum.
- 06.27 No unrestricted projection-valued spectral measure is established; the missing countable join forbids assigning the join of every countable orthogonal family, not all spectral measures for particular operators nor alternative coefficientwise summation rules; physical interpretations would need further choices, and no empirical claim is made.
- 06.28 No unrestricted Riesz, projection, countably additive spectral-measure, or field-valued least-operator-norm theorem is asserted.
- 06.29 The further directions are extensions, not claimed resolutions of named problems; closed residue is not asserted to settle every orthoclosed subspace.
- 06.30 The repository comparison used READMEs, article openings and stated labels; it is not a claim to have checked every theorem of the 99-page spectral report, every manuscript or every Lean declaration, and it does not override the distinctions of `docs/FORMALIZATION.md`.
- 06.31 No first non-Archimedean Hilbert space, first failure of Riesz representation or first non-Archimedean spectral theorem is claimed; Ishizuka’s setting is not automatically comparable.
- 06.32 Priority is to be assessed at the level of the precise hypotheses and combined classification; independent review and a wider bibliographic check remain necessary.
- 06.33 The identified inputs (Baire, closed graph, bounded inverse, Riesz, closed-operator adjoints, positive square roots, Hahn–Neumann support calculus) are not proved from set theory; class replacement and normal forms are used only globally; no assertion rests on numerical evidence; the finite checks do not verify unbounded domains, infinite-dimensional boundedness, Baire category, Hahn support arguments, proper-class reasoning or the nonexistence of projections, are not a substitute for the proofs and are not a Lean formalization; successful typesetting does not prove the mathematics.

B.3 The merge (6 items)

- M.1 [Theorem 8.10](#) is stated for closed linear relations and scalars of positive valuation only; it gives neither the finite-scalar clause of [Theorem 8.3](#) nor the orthogonal complement in [Theorem 8.7](#).
- M.2 [Proposition 6.7](#) concerns a subspace that is not orthoclosed; the orthoclosed case of [Question 15.2](#) is open.
- M.3 The comparison with `ihs:fr:thm:fredholm` recovers only its determinant-free clauses.
- M.4 The spectral report’s cluster-rotation question (`ihs:fr:sec:questions`) receives only necessary conditions; it and the other questions of [Section 15.3](#) remain open.
- M.5 Results printed by citation are not reproved here; their proofs are those of the cited reports, which are themselves unrefereed drafts.
- M.6 Nothing in this report is formalized in Lean, and `docs/FORMALIZATION.md` maps none of its labels.

C A dependency and hypothesis checklist

Table 4: Hypotheses and proof mechanisms of the main statements.

Statement	Essential hypotheses	Proof mechanism
Inner product and norms	Set support; ordinary positive Hilbert form	Leading coefficient and binomial root
Spherical completeness	Set-sized group and nested set of balls	Compatible fixed truncations
Automatic adjoints	Everywhere-defined adjoint; ordinary complete coefficient spaces	Coefficient closed graph, Baire, separation by constants
Projection normal form	Self-adjoint idempotent; ordinary Hilbert coefficients	Integral leading projection and direct rotation
Unitary group, finite straightening	Adjointable unitary; finite orthogonal family	Residue, formal exp/log, polar normalization
Infinite graph threshold	Ordinary bounded T ; scalar of negative valuation	Residue $\ker T \oplus \text{ran } T$ and closed-range inverse
Infinitesimal unbounded graphs	Closed densely defined A ; monomial $q = t^\eta$, $\eta > 0$	Coefficientwise adjoint domain; $A^{**} = A$; residue $\text{Dom } A \oplus 0$
Scaled closed relations	Closed relation; $v(b) > 0$	Residue $\text{dom } R \oplus \text{mul } R$; closed graph theorem for the operator part
Lattice dichotomy	Nontrivial group; ordinary Hilbert coefficient space	Three-coordinate graph intersection; all lines split
Positive kernel, missing join	Nontrivial group; ℓ^2	Residue of $M_q(\Lambda)$; closedness of projection residues
Countable straightening	Common support; $\mathbb{Q}\eta \subseteq \Gamma$ for the second obstruction	Finitely many pairs at exponent η ; well-ordered $S + S$
Metric rigidity	Strictly positive ordinary residue; $H \neq 0$	Automatic adjoint of a bijective isometry; ordinary bounded inverse
Fredholm least squares	Ordinary closed range; finite kernel and cokernel; positive-order global perturbation	Schur reduction, finite orthogonal complements, Penrose compression
Exact inverse scale	Nonzero finite Schur rank	Integral diagonal reduction and a forced inverse pivot
Scalar extension	Ordered group embedding; no cofinality needed	Transport of finite algebra and admissible strong sums
Full-class adjoints	Set coefficient spaces; class replacement; everywhere-defined adjoint; no global choice	Set image of constants, then the automatic adjoint proof
Fine discreteness	Set index and all surreal radii	Surreal cut below a set of distances

No theorem in the main set-sized treatment assumes a divisible group. No infinite-dimensional theorem uses a finite matrix spectral theorem to diagonalize an arbitrary Hahn operator. No unqualified exchange of an ordinary Hilbert coordinate sum with a strong Hahn family is performed. All nearest-point statements compare exact ordered-field norms; they do not replace minimization by an unproved infimum.

D Verification suites

Two finite suites accompany the report, byte-identical to the deliveries under prefixed names; neither is a proof of any infinite-dimensional, Baire, domain, lattice or class assertion.

- `code/03-orthogonal-splitting-checks.py` (Python with SymPy) runs 16 named exact symbolic checks: graph-projection idempotence and self-adjointness; the direct-rotation identities $W^*W = WW^* = I - (P - P_0)^2$ and $WP_0 = PW$, and unitarity and conjugation through order q^8 ; the Schur identity $LAR = \text{diag}(B, S)$; the four Penrose equations for a complex rational rank-two matrix; the pole example of [Example 13.4](#) (two inverse identities and $v(A^{-1}) = -4\eta$); and the triangular inverse-valuation formula on 125 positive exponent triples. Its recorded run is `data/03-orthogonal-splitting-check_results.json` (the `.txt` file is a byte copy); `data/03-orthogonal-splitting-build_report.txt` records the build of the source’s own PDF, which is not shipped.
- `code/06-hilbert-foundations-verify_finite.py` (standard-library Python with exact fractions) runs 234 identity checks in six groups: rank-one projections (60), the two-projection intertwiner before normalization (60), graph projections (60), graph intertwiners (40), the positive block operator (12), and formal prefixes to order 14 (2). Its recorded run is `data/06-hilbert-foundations-verification.json`. `code/06-hilbert-foundations-build.sh` is the source’s build script for its own manuscript, which is not shipped; it is kept as delivered.

Both suites were rerun for this report on copies (Python 3.14.4, SymPy 1.14.0): 03 printed “16 checks passed, including 125 exponent triples”, and its written JSON equals the recorded one except for the recorded Python version (3.13.5); 06 printed status PASS with 234 checks, and its written JSON equals the recorded one up to line endings. Both scripts write their outputs (03 into a `verification` directory next to the script, 06 into `verification.json` in the working directory by default), so they should be run on a copy; the README gives the commands.

References

- [1] J. Aguayo and M. Nova, *Non-Archimedean Hilbert like spaces*, Bulletin of the Belgian Mathematical Society–Simon Stevin **14** (2007), no. 5, 787–797. doi:10.36045/bbms/1197908895.
- [2] K. E. Avrachenkov and J.-B. Lasserre, *Analytic perturbation of generalized inverses*, Linear Algebra and its Applications **438** (2013), no. 4, 1793–1813. Author-deposited record: <https://inria.hal.science/hal-00926609>. Coauthor’s description: <https://www.laas.fr/fr/hompages/lasserre/softwares/singular-perturbations/>.
- [3] V. Bagayoko, L. S. Krapp, S. Kuhlmann, D. Panazzolo, and M. Serra, *Automorphisms and derivations on algebras endowed with formal infinite sums*, preprint, 2024. <https://arxiv.org/abs/2403.05827>.
- [4] H. Bursztyn and S. Waldmann, *Deformation Quantization of Hermitian Vector Bundles*, arXiv:math/0009170, 2000. <https://arxiv.org/abs/math/0009170>.
- [5] J. H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001. ISBN 978-1-56881-127-7. <https://www.routledge.com/On-Numbers-and-Games/Conway/p/book/9781568811277>.
- [6] P. Freni, *On vector spaces with formal infinite sums*, Applied Categorical Structures **34**, article 15 (2026). <https://doi.org/10.1007/s10485-025-09844-w>.
- [7] S. S. Holland, Jr., *Orthomodularity in infinite dimensions; a theorem of M. Solèr*, Bulletin of the American Mathematical Society **32** (1995), 205–234. <https://doi.org/10.1090/S0273-0979-1995-00593-8>. Preprint: <https://arxiv.org/abs/math/9504224>.
- [8] K. Ishizuka, *A spectral theorem for a non-Archimedean valued field whose residue field is formally real*, Extracta Mathematicae **39**(2) (2024), 235–254; online publication January 3, 2025. <https://doi.org/10.17398/2605-5686.39.2.235>.
- [9] T. Kato, *Perturbation Theory for Linear Operators*, second edition, Classics in Mathematics reprint, Springer, 1995. <https://doi.org/10.1007/978-3-642-66282-9>.
- [10] S. Lack and S. Tobin, *A characterisation for the category of Hilbert spaces*, Applied Categorical Structures **33**, article 13 (2025). <https://doi.org/10.1007/s10485-025-09805-3>.
- [11] J. Müller and A. Strohmaier, *The theory of Hahn meromorphic functions, a holomorphic Fredholm theorem and its applications*, Analysis & PDE **7** (2014), no. 3, 745–770. doi:10.2140/apde.2014.7.745; <https://arxiv.org/abs/1205.0236>.
- [12] B. H. Neumann, *On ordered division rings*, Transactions of the American Mathematical Society **66** (1949), 202–252.
- [13] R. Penrose, *A generalized inverse for matrices*, Proceedings of the Cambridge Philosophical Society **51** (1955), 406–413.
- [14] B. Simon, *Unitaries Permuting Two Orthogonal Projections*, arXiv:1703.05437, 2017, version 2. <https://arxiv.org/abs/1703.05437>. Formula (4) records the classical direct rotation.
- [15] B. Simon, *Tosio Kato’s work on non-relativistic quantum mechanics: part 1*, Bulletin of Mathematical Sciences **8** (2018), 121–232, especially the discussion of two projections. <https://doi.org/10.1007/s13373-018-0118-0>.

- [16] M. P. Solèr, *Characterization of Hilbert spaces by orthomodular spaces*, Communications in Algebra **23** (1995), 219–243. <https://doi.org/10.1080/00927879508825218>.
- [17] Surreal project, *Infinite-Dimensional Hahn Spectral Theory*, AI-assisted research report, directory docs/surcomplex/infinite-dimensional-hahn-spectral-theory, label prefix ihs;; read at the sources' pins 9a385d3 and a45d722, unchanged at the placement a4dcb91.
- [18] Surreal project, *Three Duals at Surreal Scales: Strong Hahn Operators, Completion Defects, and Invisible Functionals*, AI-assisted research report, directory docs/surcomplex/three-duals-of-hahn-vector-spaces, label prefix duals;; same pins.
- [19] Surreal project, *Spectral theory* (finite matrices over real closed fields and Hahn workspaces), AI-assisted research report, directory docs/surcomplex/spectral-theory; labels thm:leastssquares, thm:scales, lem:DeltaInvariant, spec:thm:svd.
- [20] Surreal project, *Analysis* report, directory docs/surcomplex/analysis; label a:thm:discrete.
- [21] Surreal project, *Foundations* report and the collection's notation and reading guide, directory docs/foundations-and-computation/foundations; label found:thm:discrete.